SEC-FRONT · Handbook Orientation · Front

Industrial Instrumentation Field Handbook

This handbook establishes a vendor-neutral foundation for interpreting industrial measurement and actuation equipment. It addresses temperature, pressure, flow, final elements, level, electrochemical measurement, machinery measurement, and transmitter power arrangements. The scope includes operating principles, representative specifications, installation considerations, calibration evidence, and diagnostic reasoning.

Each technical section contains six pages in a fixed sequence: definition, principle, specification, installation, calibration, and diagnostics. The definition identifies the measurand or controlled function. The principle explains the physical conversion. The specification separates representative values from application limits. The remaining pages address implementation, measurement evidence, and fault discrimination.

The glossary establishes canonical names and abbreviations. Section identifiers remain stable across the six page types. The comparison matrix consolidates paired technologies; the final appendix records calibration and unit-conversion conventions. Numerical examples describe plausible configurations rather than universal acceptance criteria or procurement guarantees.

Field work requires competent personnel, approved isolation, verified absence of hazardous energy, and the applicable equipment instructions. This handbook supplies training context; it does not authorize live intervention, override protective functions, or replace pressure, electrical, chemical, or machinery safety requirements. Calibration adjustments require recorded authorization and preserved as-found evidence.

Read the application requirements before interpreting a numerical limit. Distinguish sensor performance from complete-system performance, and distinguish calibration from adjustment or functional testing. A credible result identifies the measurand, reference conditions, measurement uncertainty, and decision rule. Installation effects remain relevant even when laboratory calibration is satisfactory.

SEC-GLOSS · Canonical Glossary · Front

Canonical titles identify sections; abbreviations identify devices in prose.

Table SEC-GLOSS-1: Canonical terms and abbreviations.

Canonical term	Abbreviation
Thermocouple Temperature Sensors	TC
Resistance Temperature Detectors	RTD
Pressure Transmitters	PTX
Differential Pressure Transmitters	DPTX
Magnetic Flow Meters	MAG
Vortex Flow Meters	VOR
Control Valves	CV
Solenoid Valves	SV
Guided Wave Radar Level Transmitters	GWR
Capacitive Level Switches	CAP
pH Sensors	pH
ORP (Redox) Sensors	ORP
Piezoelectric Vibration Sensors	PIEZO
Eddy Current Proximity Probes	EDDY
Loop-Powered Two-Wire Transmitters	2W
Four-Wire Transmitters	4W
Measurand	Quantity measured
Calibration	Reference comparison
Adjustment	Response modification
Measurement uncertainty	Result dispersion
Metrological traceability	Documented calibration chain
As-found result	Before adjustment
As-left result	After intervention
International System of Units	SI

Temperature: °C or K; pressure: Pa; flow: m^3/s; length: m.
Electrochemical potential: V; acceleration: m/s^2; current: A.
pH and relative permittivity are dimensionless quantities.

SEC-TC · Thermocouple Temperature Sensors · Definition

The canonical abbreviation is TC. The device produces a thermoelectric voltage from two dissimilar conductors subjected to a temperature difference. A measuring junction experiences the process temperature; the reference junction occurs where the thermoelectric conductors connect to the measuring circuit. Absolute temperature requires knowledge of the reference-junction temperature and the applicable voltage–temperature relationship.

The measurand is temperature at the measuring junction, not the surrounding pipe temperature or the average temperature of the entire sheath. Heat transfer, immersion, and junction construction influence the relationship between indicated and actual process temperature. A protective assembly can withstand the application while introducing a substantial response delay.

Type designation identifies the conductor pair and reference function. Type K uses nickel-based alloys and serves many general industrial applications. Type T uses copper and a copper–nickel alloy and suits lower-temperature service. Selection also considers atmosphere, conductor diameter, insulation, and expected exposure duration.

The sensing junction requires no excitation current. A connected transmitter or indicator still requires electrical power, input protection, linearization, and reference-junction compensation. Grounded junctions couple electrically to the sheath; ungrounded junctions provide isolation at the cost of a different thermal response.

Calibration evidence applies to the identified assembly and conditions. Replacing cable, altering termination materials, or changing insertion geometry can introduce errors beyond the calibration result. The device is therefore interpreted as a temperature-sensitive electrical system rather than an interchangeable length of wire.

This device must not be confused with Resistance Temperature Detectors (section SEC-RTD), which infer temperature from an electrically excited resistance element.

SEC-TC · Thermocouple Temperature Sensors · Principle

TC operation follows the Seebeck effect. The circuit voltage depends on the thermoelectric properties of both conductors integrated over their temperature gradients. The junction provides electrical continuity; it is not the sole location where the voltage originates.

For a homogeneous conductor pair, the measured voltage corresponds to the difference between the reference-function voltages at the measuring and reference temperatures. Electronic compensation measures the terminal temperature, determines its equivalent reference-function voltage, and combines that value with the measured input before inverse conversion.

A type K circuit with a measuring junction at 100 °C and reference junction at 0 °C produces approximately 4.096 mV. Its local sensitivity near room temperature is approximately 41 µV/°C. Neither value establishes a constant sensitivity over the complete operating range.

Intermediate metals do not introduce a net thermoelectric error when their paired transitions remain at the same temperature. Unequal terminal temperatures defeat this cancellation. Extension conductors must therefore match the approved type and temperature range, and terminal assemblies must minimize thermal gradients.

Conductor inhomogeneity changes the voltage when the affected region lies within a temperature gradient. Moving an aged assembly can alter its indication even when the measuring junction remains unchanged. Electrical continuity alone cannot demonstrate thermoelectric integrity.

Input filtering reduces electrical noise but does not correct thermal conduction error. Reference compensation, linearization, insulation integrity, and stable insertion conditions contribute separately to the final temperature result. Their contributions require distinct evidence during verification.

SEC-TC · Thermocouple Temperature Sensors · Specification

The following values describe a representative type K assembly. They are not simultaneous ratings for every sheath, junction, or cable construction. The procurement specification identifies conductor type, tolerance class, sheath material, diameter, junction form, insertion length, and termination arrangement.

Table SEC-TC-1: Representative type K characteristics.

Property	Representative value
Selected service interval	−40 to 1000 °C
Reference output at 100 °C	4.096 mV at 0 °C reference
Local sensitivity near 25 °C	Approximately 41 µV/°C
Example sheath diameter	6 mm
Compensation; Resistance Temperature Detectors	Reference at 0 °C or measured °C

The comparison property identifies compensation required for this topic; the sibling element does not require that thermoelectric correction. Type K reference values describe the standardized curve, not an individual assembly's measured error.

Nickel-alloy sheaths suit many oxidizing high-temperature applications, but chemical compatibility requires the actual process composition. Insulation resistance, vibration exposure, and thermal cycling require separate limits. A favorable temperature rating does not establish pressure containment adequacy.

The complete uncertainty includes element deviation, reference compensation, input measurement, and installation effects. State whether an error limit refers to degrees Celsius, percent of reading, or percent of configured span. Response-time claims identify the medium, flow condition, and percentage response criterion.

SEC-TC · Thermocouple Temperature Sensors · Installation

Installation places the TC measuring junction in a representative thermal region while maintaining the approved pressure boundary. The assembly drawing defines insertion depth, process connection, protective geometry, and support. No universal immersion ratio replaces a thermal-conduction assessment for the actual application.

The termination schedule preserves conductor polarity and type from the sensing assembly to the receiving circuit. Extension and compensating conductors have specific temperature limits. Ordinary copper cable is appropriate only after the designated reference transition and compensation arrangement.

Terminal boxes require protection against moisture and unequal heating. The reference-junction sensor must represent the temperature of the thermoelectric-to-copper transitions. A nearby hot enclosure surface or sunlit terminal strip can create a compensation error despite correct wiring.

Shielding and grounding follow the documented electrical design. A grounded measuring junction changes the available isolation path and can expose the input to process common-mode voltage. Cable routes separate sensitive millivolt circuits from switching and variable-frequency power conductors.

Mechanical installation avoids unsupported vibration and excessive bending of mineral-insulated assemblies. Protective pockets require pressure and flow-induced vibration assessment by the responsible designer. Their suitability is not established by the sensing element's temperature rating.

Commissioning records the insertion position, conductor identification, polarity, reference-compensation mode, and configured type. A stable process comparison evaluates representative temperature and response lag. Any insulation test uses the approved method with sensitive electronics disconnected and the assembly safely isolated.

SEC-TC · Thermocouple Temperature Sensors · Calibration

A TC converts the thermal difference between a process junction and a reference transition into a small electrical voltage. The two conductors have different thermoelectric characteristics. Recovering the process temperature requires both the measured voltage and the reference temperature, interpreted through the designated conductor-pair function.

A thermal calibration compares the complete sensing assembly with a suitable reference in a stable bath or furnace. The selected points bracket the intended service interval where practicable. Equal immersion and adequate stabilization reduce differences caused by thermal gradients.

Electrical simulation evaluates the receiving circuit separately. The calibrator's compensation mode and the receiver's compensation mode must form a defined reference arrangement. An accidental combination of two compensations, or omission of both, can produce a convincing but invalid result.

An illustrative check uses 0, 100, and 400 °C with ascending and descending observations. Actual points follow the application and reference capability. The record includes reference correction, stability, immersion, input voltage, indicated temperature, and measurement uncertainty.

As-found results precede any adjustment. A failed element comparison is not corrected by changing transmitter scaling without investigating conductor degradation or installation error. Adjustment addresses an identified component and retains the original evidence.

As-left verification repeats the relevant points and confirms input type, compensation, and output scaling. Metrological traceability belongs to the reported result through documented reference comparisons. A continuity check or simulated input alone does not establish calibration of the sensing assembly.

SEC-TC · Thermocouple Temperature Sensors · Diagnostics

TC diagnosis separates electrical faults, reference-compensation faults, and thermal errors. An open conductor commonly drives the receiver toward its configured failure indication. That indication documents the receiver's behavior and does not prove the location of the discontinuity.

A reversed conductor pair can produce an indication that moves opposite to a temperature rise. Incorrect extension material may create errors dependent on terminal or cable temperature. Record the environmental dependence instead of assuming a fixed offset.

A reading that changes when terminal-box temperature changes suggests a reference-transition or compensation problem. Check the configured type, terminal temperature measurement, conductor transitions, and spatial heating under an approved diagnostic procedure.

Intermittent spikes may arise from loose connections, insulation breakdown, common-mode disturbance, or vibration-induced conductor damage. Comparison of the raw input with the processed indication helps distinguish electrical interference from excessive filtering or software scaling.

A slow yet stable response can result from protective geometry, poor thermal contact, or insufficient immersion. Lengthening the damping time does not repair the underlying thermal coupling. Process stratification can also make two correctly operating assemblies disagree.

Drift after high-temperature exposure or relocation warrants assessment of conductor inhomogeneity. Replacement decisions use recorded service history and comparison evidence. Restore the approved configuration, document the diagnosed mechanism, and retain the as-found result before returning the channel to service.

SEC-RTD · Resistance Temperature Detectors · Definition

The canonical abbreviation is RTD. The device infers temperature from the resistance of a metallic sensing element. A measuring circuit supplies excitation and determines resistance from the resulting electrical response. Platinum elements provide a standardized relationship suitable for stable industrial temperature measurement.

A Pt100 element has nominal resistance of 100 Ω at 0 °C. The designation does not independently establish tolerance class, lead arrangement, construction, or service interval. Those attributes belong to the specified assembly.

Wire-wound and thin-film constructions differ in mechanical behavior and permissible temperature exposure. The element sits within an insulated assembly that transfers process heat to the sensitive material. The sheath and protective pocket influence response time and conduction error.

Two-, three-, and four-lead arrangements address lead resistance differently. The lead count describes the sensing circuit, not the transmitter power arrangement. Excitation produces self-heating, so apparent precision can exceed the validity of the thermal result.

This device must not be confused with Thermocouple Temperature Sensors (section SEC-TC), which generate a thermoelectric voltage dependent on measuring and reference temperatures.

SEC-RTD · Resistance Temperature Detectors · Principle

RTD resistance increases with temperature over the intended platinum operating interval. The standardized relationship is nonlinear. Conversion therefore uses the appropriate reference equation rather than a single constant resistance change per degree.

For a common standardized Pt100 curve, resistance at 100 °C is approximately 138.505 Ω. Near 0 °C, sensitivity is approximately 0.391 Ω/°C. These values describe the nominal function and not the deviation of an individual element.

In a two-lead circuit, measured resistance includes both lead resistances. A three-lead circuit compensates under assumptions about lead matching and measuring topology. Four-lead measurement separates excitation and voltage sensing, greatly reducing lead-resistance contribution when implemented correctly.

Self-heating depends on excitation power and heat dissipation into the medium. A circuit using 1 mA through 100 Ω dissipates 0.1 mW in the element. The resulting temperature rise remains installation-dependent.

The complete measurement involves resistance determination, curve conversion, lead treatment, and thermal coupling. Stable electronics do not remove errors caused by mechanical strain, inadequate immersion, or changes in element properties.

SEC-RTD · Resistance Temperature Detectors · Specification

Specify the platinum curve, nominal resistance, tolerance class, construction, and lead configuration. Element tolerance and complete-channel accuracy are different requirements. The service interval must lie within the construction-specific limits.

Table SEC-RTD-1: Representative platinum assembly.

Property	Representative value
Selected operating interval	−100 to 400 °C
Nominal resistance at 0 °C	100 Ω
Nominal resistance at 100 °C	138.505 Ω
Illustrative excitation	0.5 mA
Compensation; Thermocouple Temperature Sensors	Lead resistance in Ω; no junction °C correction

The comparison row identifies the correction relevant to this topic. Three-lead compensation requires sufficiently matched leads; four-lead measurement requires the correct terminal topology.

A common class A tolerance expression is ±(0.15 + 0.002 × absolute temperature) °C, within the applicable standard and construction interval. At 100 °C this expression gives ±0.35 °C for the element, not for every installed channel.

State insulation, sheath compatibility, vibration limits, and response conditions separately. Stainless steel may suit the example assembly, but material selection depends on corrosion, pressure-boundary design, and process cleanliness requirements.

SEC-RTD · Resistance Temperature Detectors · Installation

The RTD assembly occupies a location representative of the required process temperature. The mechanical design controls pressure integrity, immersion, support, and maintenance access. Thermal conduction along the sheath requires assessment where the process differs greatly from ambient temperature.

Terminal identification follows the assembly drawing rather than assumptions based only on wire color. Three-lead circuits require the correct pair allocation; four-lead circuits maintain separate excitation and sense connections.

Lead routes minimize electrical interference and temperature-dependent mismatch. Extension connections use suitable copper conductors and stable terminals. Moisture ingress can create leakage paths that produce low or unstable indications.

Excitation settings remain compatible with the element and thermal environment. Low-flow gas applications can exhibit more self-heating than a well-agitated liquid at the same temperature. A change of measuring circuit therefore requires review.

Commissioning records element designation, curve, lead topology, insertion geometry, and receiver configuration. Any resistance or insulation verification occurs under approved isolation and with sensitive electronics protected from inappropriate test voltages.

SEC-RTD · Resistance Temperature Detectors · Calibration

RTD calibration compares indicated temperature or measured resistance with a reference at stabilized temperatures. The record identifies whether the sensing assembly, receiver, or complete channel is under evaluation.

A bath comparison requires appropriate immersion and sufficient separation from heater gradients. At low and moderate temperatures, a stirred liquid bath can provide better uniformity than an unsuitable dry block. Reference capability and process cleanliness constrain the choice.

Resistance simulation verifies receiver conversion and lead handling without proving the element's thermal performance. The simulator reproduces the configured lead topology. Adding an undefined lead resistance can invalidate the result.

An illustrative sequence evaluates −20, 50, and 150 °C. Record ascending and descending readings, reference corrections, stability, excitation, and measurement uncertainty. Point selection follows the required service interval rather than this example.

Preserve as-found results and investigate self-heating or insertion differences before adjustment. As-left verification confirms the nominal curve and terminal topology. Calibration coefficients apply only to the identified assembly and justified interpolation interval.

SEC-RTD · Resistance Temperature Detectors · Diagnostics

RTD faults include open elements, shorted leads, leakage, and incorrect lead allocation. The receiving circuit's failure policy determines the resulting indication. Failure direction is therefore a configuration attribute, not a universal device property.

A persistent positive error in a two-lead circuit may correspond to added lead resistance. An ambient-dependent error in a three-lead circuit may indicate unequal lead resistance or unequal terminal conditions.

A low reading accompanied by reduced insulation resistance suggests leakage across the element or cable. Moisture at a terminal box can create a parallel path without a complete short circuit.

Excitation-dependent temperature changes suggest self-heating. Compare only under controlled thermal conditions; process changes can otherwise imitate the same pattern. Mechanical strain or thermal cycling may produce irreversible element drift.

Separate the element, lead circuit, and receiver through authorized reference comparisons. Do not conceal a lead or leakage fault by applying a software offset. The diagnostic record identifies the fault mechanism and verifies recovery under the original service configuration.

SEC-PTX · Pressure Transmitters · Definition

The canonical abbreviation is PTX. The device converts pressure at a process connection into an electrical measurement signal. A sensing assembly responds relative to a defined reference, and electronics convert that response into engineering units and a configured output. The pressure reference is integral to the meaning of the result.

Gauge measurement uses atmospheric pressure as its reference. Absolute measurement uses an evacuated reference. A sealed-gauge construction uses a sealed reference established during manufacture and does not continuously follow atmospheric changes.

The process diaphragm, isolation fill, and sensing element need not be the same component. Remote seals separate the sensing assembly from difficult process conditions but add thermal and elevation effects to the measurement.

Configured span describes the output interval; it does not redefine the pressure boundary's maximum rating. Overpressure capability, process connection rating, and material compatibility require independent review.

This device must not be confused with Differential Pressure Transmitters (section SEC-DPTX), which measure the pressure difference between two process pressure connections.

SEC-PTX · Pressure Transmitters · Principle

PTX sensing converts diaphragm deformation into a measurable electrical change. Common constructions use strain-sensitive resistance or capacitance. The diaphragm geometry, reference cavity, and temperature compensation establish the usable measurement interval.

An isolation diaphragm transfers process pressure through a fill fluid. Its compliance and the fill properties introduce effects not present in an ideal direct pressure connection. Temperature change can alter fill volume and contribute to zero displacement.

A linear 0 to 1 MPa configuration associates 0 MPa with 4 mA and 1 MPa with 20 mA. At 0.5 MPa the ideal output is 12 mA. This mapping describes output scaling rather than the sensing principle.

Pressure varies with elevation in a liquid-filled connection. For water of density 1000 kg/m^3, a 0.2 m elevation difference corresponds to approximately 1.96 kPa under standard gravity. The sign depends on the relative elevations.

Digital compensation reduces characterized temperature effects but cannot infer an unknown blocked connection or wrong reference type. The conversion chain therefore includes physical pressure transmission as well as electronics.

SEC-PTX · Pressure Transmitters · Specification

Define pressure reference before specifying range or error. The representative configuration below describes a gauge channel. Values remain subject to the selected sensing assembly and process connection rating.

Table SEC-PTX-1: Example gauge-pressure channel.

Property	Example requirement
Configured gauge interval	0 to 1 MPa
Illustrative reference error limit	±0.1 % of 1 MPa span
Terminal supply example	24 V
Ambient design interval	−20 to 60 °C
Reference; Differential Pressure Transmitters	Atmosphere; 0 kPa gauge at vented input

The comparison row distinguishes the atmospheric reference used here from a second process pressure input. Absolute and sealed-gauge versions require different reference statements.

Wetted stainless steel is a possible construction, not a universal compatibility guarantee. Diaphragm alloy, seals, fill fluid, and connection geometry require the process composition and cleaning conditions.

List temperature effects, long-term stability, overload rating, and response separately from reference accuracy. An output span narrowed below the nominal sensing interval can increase the percentage-of-span contribution of fixed errors.

SEC-PTX · Pressure Transmitters · Installation

PTX installation preserves the pressure boundary and presents representative pressure at the sensing connection. Approved drawings define isolation, connection materials, supporting structure, and access. Personnel verify safe isolation before any connection is opened.

The mounting elevation becomes part of the measurement for liquid-filled connections. Record the process tapping and sensing elevations. A change in location can alter the zero even when the transmitter calibration remains unchanged.

Connections avoid unintended liquid accumulation in gas service and trapped gas in liquid service, subject to the engineered arrangement. Thermal protection for hot service follows the approved design rather than an improvised length of tubing.

Gauge-reference vents remain protected from blockage and environmental contamination. Enclosure sealing does not justify sealing an atmospheric reference path intended to breathe.

Commissioning verifies the pressure reference, units, configured endpoints, failure output, and receiving-channel scaling. Leak testing and pressure-boundary acceptance follow authorized procedures; an electrical output check does not establish mechanical integrity.

SEC-PTX · Pressure Transmitters · Calibration

A PTX interprets pressure delivered through its process connection and converts the sensing response into a reported value and electrical output. The chosen reference determines whether that value represents pressure relative to atmosphere, an evacuated cavity, or a sealed datum. Output interpretation is incomplete without this reference information.

Calibration compares the pressure input with a suitable reference while observing both the indicated pressure and output current. The reference uses compatible pressure media and a range that establishes adequate measurement uncertainty.

For the 0 to 1 MPa example, five points at 0, 0.25, 0.5, 0.75, and 1 MPa reveal endpoint and intermediate behavior. Descending observations evaluate hysteresis. Applied pressure stability and elevation correction enter the result.

Gauge-zero evaluation requires a defined atmospheric condition. Absolute-zero evaluation requires appropriate absolute reference capability; venting an absolute instrument does not create zero absolute pressure.

Preserve as-found data before sensor adjustment or output adjustment. These operations address different conversion stages. As-left verification confirms pressure indication, output mapping, and the original reference conditions.

SEC-PTX · Pressure Transmitters · Diagnostics

PTX diagnosis compares physical pressure transmission, sensing response, and output interpretation. A frozen reading can originate from a blocked process connection even when the electronics report no internal fault.

A constant offset may result from mounting elevation, reference-vent obstruction, or an incorrect gauge-versus-absolute selection. Relate the offset to atmospheric and temperature changes before assigning it to sensor drift.

A stable local pressure indication with an incorrect receiving indication suggests output scaling, current measurement, or channel configuration. Confirm the same units and endpoints at both ends of the signal path.

Erratic response can reflect process pulsation, trapped gas, loose terminals, or supply limitations. Damping can obscure the symptom while leaving its physical cause unchanged.

Damage after overload requires assessment against the rated boundary and measurement characteristics. Do not pressurize beyond an approved test plan to reproduce a fault. Record the cause, intervention, and as-left comparison before release.

SEC-DPTX · Differential Pressure Transmitters · Definition

The canonical abbreviation is DPTX. The device measures the pressure difference between designated high-side and low-side process connections. Its primary measurand is high-side pressure minus low-side pressure. Flow, level, and filter loading are derived applications rather than direct measurements of those quantities.

Static pressure acts on both sides and can greatly exceed the differential interval. Differential measurement capability does not establish the permissible common pressure or one-sided overload.

Impulse connections communicate pressure from the process to the sensing assembly. Their contents, elevations, and temperatures influence the measured difference. Unequal liquid heads can create an apparent differential with no corresponding process change.

Square-root conversion applies only where the application relationship requires it, such as an appropriate restriction-based flow calculation. A level or filter application commonly requires linear pressure interpretation.

This device must not be confused with Pressure Transmitters (section SEC-PTX), which use a defined atmospheric, absolute, or sealed reference instead of two process pressure inputs.

SEC-DPTX · Differential Pressure Transmitters · Principle

DPTX sensing responds to the imbalance between two pressures. Isolation diaphragms and fill systems transfer each pressure to the measuring element. Mechanical design allows a small differential to be resolved while common pressure loads the assembly.

For restriction-based flow under defined conditions, differential pressure varies approximately with the square of flow. At one-quarter of the full differential, the corresponding idealized flow is one-half of full flow. Density and restriction geometry remain part of the calculation.

Hydrostatic level interpretation follows liquid density, gravity, and vertical height. A vapour-space connection can cancel vessel pressure while a reference liquid head introduces a fixed or temperature-dependent contribution.

Common-pressure effects and differential effects are distinct. Equal pressures at both inputs define zero differential but do not mean the assembly is unpressurized. This distinction governs safe isolation and test planning.

The electronics can linearize or apply application calculations, but cannot correct unknown impulse-line composition. Valid measurement therefore requires both a defined physical connection model and the corresponding configured transfer function.

SEC-DPTX · Differential Pressure Transmitters · Specification

The differential interval, static-pressure capability, and unilateral overload capability require separate entries. The following example addresses a low differential at appreciable common pressure.

Table SEC-DPTX-1: Illustrative differential channel.

Property	Illustrative value
Configured differential interval	0 to 25 kPa
Example common-pressure rating	10 MPa
Reference-condition error target	±25 Pa
Configured damping example	1 s
Input reference; Pressure Transmitters	Second process input; 0 Pa when equal

The 25 Pa error target corresponds to 0.1 % of the example differential span. It does not include every common-pressure, ambient-temperature, or installation contribution.

Specify fill fluid, diaphragm material, manifold rating, and required pressure connections. A process-compatible diaphragm is insufficient if the manifold seals or impulse tubing are incompatible.

Remote-seal elevation and temperature effects require a documented application calculation. The configured lower value can be negative because of reference head; the electrical zero is not necessarily physical zero differential.

SEC-DPTX · Differential Pressure Transmitters · Installation

DPTX installation maintains the designated high-side and low-side pressure paths. The engineered drawing establishes tapping locations, elevations, manifold arrangement, and the intended contents of each impulse connection.

Gas, liquid, and condensing service require different arrangements. The objective is a stable, known pressure-transmission system rather than a generic tubing slope. Heat tracing and insulation require design review where freezing or condensation changes the reference head.

Both connections require compatible pressure ratings and materials. Unequal thermal exposure can create unequal density or seal-fill expansion effects. Support minimizes vibration and stress on the process connections.

Manifold operation follows the approved equipment-specific sequence. Equal pressure does not establish absence of stored energy; opening a vent without verified isolation can release hazardous process material.

Commissioning records side allocation, elevations, reference-head assumptions, units, linear or square-root conversion, and low-flow behavior. The acceptance record distinguishes pressure-boundary verification from electrical and application-function verification.

SEC-DPTX · Differential Pressure Transmitters · Calibration

DPTX calibration applies a known differential with both pressure references explicitly defined. The test plan respects static and one-sided pressure limits and uses an approved isolated arrangement.

For a 0 to 25 kPa linear configuration, points at 0, 6.25, 12.5, 18.75, and 25 kPa exercise the span. Ascending and descending results reveal hysteresis and repeatability under stated conditions.

Equal-input zero verification evaluates the sensing assembly at zero differential. Where common-pressure performance matters, a suitable facility evaluates the relevant static condition without assuming that an atmospheric check covers it.

A flow-configured output has a nonlinear relationship to the applied differential. At 6.25 kPa in this example, ideal square-root scaling indicates 50 % flow and 12 mA, not the linear-pressure output of 8 mA.

Preserve as-found pressure and current data, reference-head corrections, and conversion settings. As-left verification includes the application function and any low-flow cutoff. Sensor adjustment does not establish calibration of a restriction or the process impulse connections.

SEC-DPTX · Differential Pressure Transmitters · Diagnostics

DPTX faults often originate outside the sensing assembly. A blocked impulse connection may trap a pressure and create a plausible but unresponsive indication. Two blocked connections can conceal genuine process changes.

A changing reference head can produce drift associated with ambient temperature, condensation, or liquid loss. Compare the observed error with the engineered elevation and density model.

Reversed pressure paths produce the wrong differential sign. A configured sign inversion may hide this physical error and complicate later maintenance. Preserve the intended side allocation.

An unexpectedly high derived flow at low differential may indicate unsuitable square-root processing, noise, or cutoff configuration. A correct pressure comparison does not prove a correct application calculation.

Investigate leaks, manifold state, connection contents, and common-pressure effects using authorized isolation. Record evidence before adjustment; a zero correction must not conceal an unstable reference system or a compromised pressure boundary.

SEC-MAG · Magnetic Flow Meters · Definition

The canonical abbreviation is MAG. The device determines volumetric flow of an electrically conductive liquid from voltage induced as the liquid crosses a magnetic field. A lined measuring tube establishes the flow passage, electrodes detect the induced signal, and electronics relate that signal to mean velocity and cross-sectional area.

The measurand is actual liquid volume flow through the measuring tube. Conversion to mass flow requires independently established density. The method does not measure gas or a nonconductive liquid merely because either occupies the same pipe.

The insulating liner prevents the metal body from shorting the measurement field. Electrode and liner materials require chemical, thermal, and abrasion compatibility. A mechanically intact lining may still be unsuitable for vacuum exposure.

A full measuring tube and suitable process grounding are essential measurement conditions. Entrained gas, stratification, and electrode contamination can compromise the result without an obvious electronics failure.

This device must not be confused with Vortex Flow Meters (section SEC-VOR), which infer velocity from the frequency of vortices shed by a flow obstruction.

SEC-MAG · Magnetic Flow Meters · Principle

MAG operation follows electromagnetic induction. In simplified geometry, induced voltage is proportional to magnetic flux density, electrode separation, and liquid velocity. Instrument characterization relates the sensed voltage to average flow under defined conditions.

For a 100 mm internal diameter and mean velocity of 2 m/s, volume flow is approximately 0.0157 m^3/s. The calculation uses actual internal area; nominal connection size alone does not define that area precisely.

Alternating or pulsed field excitation helps distinguish the flow-dependent signal from electrode offsets. The liquid still needs adequate conductivity for stable electrical coupling to the electrodes.

Process grounding establishes a suitable electrical reference between the liquid and measurement circuit. Insulating pipes may require engineered grounding accessories. Protective earthing and measurement grounding have related but nonidentical functions.

An empty or partially filled tube violates the assumed geometry. Coating that changes electrode contact can degrade the electrical measurement. Signal processing can detect some such conditions but cannot make an unsuitable medium measurable.

SEC-MAG · Magnetic Flow Meters · Specification

Conductivity, liner compatibility, and full-tube conditions govern suitability before an accuracy claim is considered. These illustrative values describe a conductive-water application, not a universal product envelope.

Table SEC-MAG-1: Representative conductive-liquid selection.

Property	Representative value
Selected velocity interval	0.3 to 10 m/s
Example conductivity minimum	5 µS/cm = 0.0005 S/m
Example bore diameter	100 mm
Illustrative flow error	±0.5 % of reading
Conductivity threshold; Vortex Flow Meters	At least 0.0005 S/m for this example

The comparison row identifies a requirement specific to the induced-voltage method. The actual minimum conductivity follows the selected instrument and medium rather than this training value.

Polytetrafluoroethylene lining and alloy electrodes are possible choices where their ratings and compatibility fit. Abrasive slurry, hot cleaning fluid, and vacuum conditions require separate assessment.

Specify low-velocity error terms, bidirectional operation, grounding, signal outputs, and empty-tube diagnostics. A reading-relative error alone can understate performance limitations near zero flow.

SEC-MAG · Magnetic Flow Meters · Installation

MAG placement maintains a full measuring tube across the required operating conditions. The engineered location avoids persistent gas accumulation and sediment exposure at electrodes. The electrode axis follows the approved mounting instructions.

Pipe supports prevent loads from distorting the measuring tube or lined flanges. Gaskets remain outside the effective bore where required, and tightening follows the lining-specific procedure. Excessive compression can damage a liner.

The liquid grounding arrangement follows pipe conductivity, lining, and cathodic-protection conditions. Grounding accessories use materials compatible with the liquid. Unreviewed bonding can introduce unwanted current paths.

Remote sensor wiring uses the approved cable and termination practice. Electrode circuits can be sensitive to cable leakage and moisture. Power and excitation wiring remain separated from sensitive input paths as designed.

Commissioning records flow direction, actual bore, grounding arrangement, conductivity, and full-tube status. Straight-run requirements follow the selected instrument and disturbance geometry; no generic allowance overrides the approved installation limits.

SEC-MAG · Magnetic Flow Meters · Calibration

A MAG translates the electrical potential created by conductive liquid moving through an imposed magnetic field into volume flow. The measuring passage, electrode spacing, and electronic conversion link the induced signal to mean velocity. The resulting flow value depends on the passage area and a valid liquid-filled measurement condition.

A wet calibration compares delivered volume or mass over measured time with the indicated volume flow. A mass reference requires a justified density conversion at the test conditions. The reference facility controls temperature, air content, and flow stability.

Illustrative test velocities of 0.5, 2, and 5 m/s exercise low, intermediate, and higher operation. The actual points reflect the service interval and reference facility capability.

A zero check requires a full tube containing stationary conductive liquid. An empty tube is not an equivalent zero condition. Internal electronic verification assesses selected components but does not replace a complete wet flow comparison.

Preserve as-found configuration, conductivity, reference uncertainty, and measured errors. Any adjustment of a calibration factor requires authorization and supporting evidence. As-left results confirm the revised factor and relevant output scaling.

SEC-MAG · Magnetic Flow Meters · Diagnostics

MAG diagnosis begins with medium conductivity and full-tube status. A healthy electronic module can produce unreliable flow when bubbles interrupt the electrode path or the liquid conductivity lies outside the approved range.

A nonzero indication at stationary full-tube conditions may indicate grounding disturbance, electrode contamination, or an inappropriate zero setting. Confirm that the liquid is actually stationary rather than relying only on commanded valve state.

Intermittent spikes can correlate with entrained gas, pump operation, cable leakage, or switching interference. Trend electrode diagnostics and process conditions together to discriminate mechanisms.

A slow gain change can arise from deposits, liner damage, or configuration changes. Compare totalized quantity with an independent material balance only when storage, density, and timing effects are sufficiently understood.

Inspect and clean only under the approved process isolation and material-compatible procedure. Retain the original calibration factor and as-found evidence. Recovery requires restored physical measurement conditions as well as acceptable electronic checks.

SEC-VOR · Vortex Flow Meters · Definition

The canonical abbreviation is VOR. The device derives flow from the frequency of alternating vortices formed downstream of a shaped obstruction. A sensor detects the periodic disturbance, and electronics convert frequency into volume flow under the characterized operating conditions.

The method can serve liquids, gases, and steam where velocity, density, and viscosity permit stable vortex formation and detection. Each medium requires its own acceptable operating envelope.

Actual volume flow differs from standard-condition gas volume or mass flow. Pressure, temperature, and fluid-property information support those conversions; the shedding frequency alone does not establish mass.

The obstruction introduces pressure loss and a minimum usable flow. Pipe vibration and disturbed inlet profiles can interfere with detection. A missing frequency at low flow is not necessarily an electrical failure.

This device must not be confused with Magnetic Flow Meters (section SEC-MAG), which require electrically conductive liquid for induced-voltage measurement.

SEC-VOR · Vortex Flow Meters · Principle

VOR frequency is approximately proportional to velocity divided by obstruction width over the characterized shedding regime. The dimensionless Strouhal number expresses this relationship. Its usable constancy depends on flow conditions and geometry.

A hypothetical obstruction width of 20 mm, velocity of 10 m/s, and Strouhal number of 0.2 give a shedding frequency of approximately 100 Hz. This example illustrates the relationship and is not a transferable calibration constant.

Reynolds number relates inertia to viscosity. Outside the qualified regime, shedding may become unstable or the frequency relationship may depart from its characterized behavior.

The detector senses mechanical or pressure variation caused by vortices. Mechanical vibration can generate competing signals, so signal discrimination and installation stiffness matter.

Mass conversion multiplies actual volume flow by density. Steam state and gas composition influence density; wet steam or changing composition can make a nominal compensation model inappropriate even when frequency detection is stable.

SEC-VOR · Vortex Flow Meters · Specification

Specify fluid state, minimum density, viscosity, expected Reynolds number, and acceptable pressure loss. The following gas-service values are illustrative and require confirmation against the selected geometry.

Table SEC-VOR-1: Example gas-service envelope.

Property	Example value
Selected gas velocity	5 to 50 m/s
Configured pulse factor	1000 pulses/m^3
Illustrative gas-flow error	±1 % of reading
Selected process temperature	20 to 250 °C
Medium restriction; Magnetic Flow Meters	Gas permitted at 5 to 50 m/s

The comparison row identifies a gas application available with this shedding-based method. Minimum usable velocity still depends on detector sensitivity and fluid density.

Stainless steel measuring bodies can suit many clean-fluid services when connection ratings and corrosion limits are satisfied. Erosion, deposits, and multiphase flow can change the obstruction condition.

Accuracy specifies actual volume unless another quantity is explicitly defined. Standard-condition volume and mass claims also include compensation inputs and the selected property model. Straight-run and vibration requirements are procurement constraints, not optional installation preferences.

SEC-VOR · Vortex Flow Meters · Installation

VOR installation provides the qualified inlet profile and mechanical support. Required straight lengths depend on upstream bends, reducers, and throttling equipment. The applicable drawing and instrument instructions establish the actual distances.

The measuring bore aligns with the pipe and gasket openings. Protruding gaskets or steps can disturb shedding. The instrument's flow-direction marking follows the intended operating direction.

Pipe support limits vibration transmitted into the detector. Additional filtering is not a substitute for correcting excessive mechanical vibration or unsupported piping.

Steam and hot-gas arrangements account for temperature exposure, condensate behavior, and access. Insulation does not cover electronics beyond the approved boundary. Pressure and temperature compensation locations follow the engineered measurement model.

Commissioning verifies pulse factor, actual-volume units, compensation basis, cutoff, and detector diagnostics. A stationary-flow baseline helps establish vibration interference, but operational tests require approved process conditions and do not bypass protective controls.

SEC-VOR · Vortex Flow Meters · Calibration

VOR calibration compares frequency or indicated volume flow against a suitable flow reference within the qualified shedding regime. The reference conditions identify fluid, temperature, pressure, density, and viscosity.

A pulse factor states pulses per unit volume; its orientation differs from a volume-per-pulse factor. The receiving system must interpret the chosen convention consistently. Dimensional checking prevents reciprocal-factor errors.

Illustrative gas test velocities of 8, 20, and 40 m/s evaluate the selected interval. Equivalent velocities in a different fluid do not necessarily reproduce the same detector performance.

Electrical frequency simulation verifies the receiving electronics and output scaling. It does not assess the obstruction, detector coupling, inlet profile, or actual flow calibration.

Preserve as-found pulse factor and compensation settings. Any revised factor requires reference results and measurement uncertainty. As-left verification includes totalization and the applicable actual-volume, standard-volume, or mass conversion.

SEC-VOR · Vortex Flow Meters · Diagnostics

VOR diagnosis distinguishes insufficient shedding, weak detector signal, vibration interference, and incorrect compensation. Zero output below the qualified minimum flow can be normal behavior.

A indicated flow during a confirmed stationary condition suggests mechanical vibration or signal-processing configuration. Compare detected frequency with machine or pipe vibration frequencies without approaching moving equipment.

Low or unstable flow indication can follow deposits on the obstruction, changed viscosity, low gas density, or multiphase conditions. The detector may remain electrically healthy.

A correct pulse frequency with wrong mass indication directs attention to pressure, temperature, density basis, or unit conversion. Standard-condition labels require explicit reference conditions.

Investigate through approved process evidence and isolated inspection. Preserve low-flow cutoff and filtering settings before changes. A repaired detector requires verification of its characterized assembly, not merely proof of electrical continuity.

SEC-CV · Control Valves · Definition

The canonical abbreviation is CV. The device modulates fluid flow by changing an internal flow passage in response to an actuator command. The body and closure element establish the hydraulic restriction, the actuator supplies movement, and a positioner may regulate travel against the demanded position. The controlled process variable responds through the surrounding system.

Demanded position, measured travel, and actual flow are different quantities. A correct travel response does not prove the intended flow response when pressure conditions or internal geometry change.

Linear and rotary constructions provide different mechanical arrangements. Inherent flow characteristic describes the restriction under defined pressure conditions; installed characteristic incorporates the connected process.

Failure position follows the actuator and system design. Loss of electrical signal, actuator power, or air pressure can have different outcomes. A generic assumption of closed-on-failure is unacceptable.

This device must not be confused with Solenoid Valves (section SEC-SV), which use an energized electromagnetic coil to change the position of a closure mechanism.

SEC-CV · Control Valves · Principle

CV movement changes effective flow area. For nonchoked incompressible flow, flow depends approximately on the capacity coefficient and the square root of pressure difference divided by density. Gas expansion, cavitation, and choking require additional models.

An actuator converts pneumatic, electrical, or hydraulic energy into stem force or shaft torque. The available force must overcome process forces, packing friction, and the required seating load without exceeding mechanical limits.

A positioner compares demanded travel with measured travel and changes actuator drive. This local feedback can reduce positioning error but does not directly measure process flow.

A linear inherent characteristic gives approximately proportional capacity with travel under constant pressure difference. An equal-percentage characteristic gives equal percentage capacity changes for equal travel increments over its defined interval.

Installed behavior depends on the distribution of pressure losses in the entire system. Oversizing can confine operation near the seat, while friction and deadband can create cycling despite an apparently adequate controller configuration.

SEC-CV · Control Valves · Specification

Selection requires minimum, normal, and maximum flow conditions together with inlet pressure, outlet pressure, temperature, and fluid properties. Travel performance and pressure-boundary suitability remain separate requirements.

Table SEC-CV-1: Illustrative modulating assembly.

Property	Illustrative value
Nominal linear travel	25 mm
Positioner demand interval	4 to 20 mA
Example actuator air supply	400 kPa gauge
Positioning error target	±0.5 % of 25 mm travel
Command function; Solenoid Valves	Continuous 0 to 25 mm modulation

The comparison row describes continuous travel control for this example; it does not claim that every coil-operated design is restricted to binary action.

Specify body and trim materials, connection rating, leakage acceptance method, characteristic, actuator sizing, and failure action. Capacity coefficients require their convention and test basis; dimensionless shorthand can obscure units.

Noise, cavitation, flashing, and erosion evaluations use the actual process cases. A specified travel error does not establish flow accuracy or acceptable shutoff performance.

SEC-CV · Control Valves · Installation

CV installation follows the engineered flow direction, actuator orientation, and support arrangement. The body must not carry unreviewed pipe loads. Maintenance clearances include actuator removal and safe access to packing and positioner components.

Cleanliness controls prevent debris from damaging the seat or restricting actuator accessories. Flushing and pressure testing follow the approved project procedure, with sensitive components protected as specified.

Pneumatic supply quality and pressure match the actuator and positioner requirements. Tubing size and length influence response. Exhaust routing respects environmental and hazardous-area requirements.

Mechanical linkage or noncontact feedback alignment follows the equipment instructions. Incorrect geometry can create nonlinear travel indication or prevent full motion. Stored spring and actuator energy remains hazardous after the command signal is removed.

Commissioning verifies demand direction, travel limits, feedback, and each designed failure condition under an authorized test plan. Stroke tests require process permission and an exclusion zone; they never rely on an assumed safe automatic response.

SEC-CV · Control Valves · Calibration

A CV regulates the available fluid passage through commanded mechanical movement. Its pressure-containing body and internal closure determine the restriction, while the actuator delivers motion and an optional positioner maintains demanded travel. The process response depends on both that restriction and the connected hydraulic system.

Travel calibration compares demand with an independent travel reference under authorized, isolated conditions. Automatic routines can cause full actuator movement and therefore require verified process safety before initiation.

For a 25 mm example, demands of 4, 8, 12, 16, and 20 mA correspond ideally to 0, 6.25, 12.5, 18.75, and 25 mm. Ascending and descending readings reveal deadband and hysteresis.

Record actuator supply pressure, linkage geometry, travel reference, and as-found settings. Adjusting endpoints does not repair packing friction, damaged trim, or inadequate actuator force.

As-left verification confirms demand direction, travel indication, limits, and the approved failure action. Functional verification of shutoff or flow characteristic requires separate suitable testing; a successful travel calibration alone does not prove either property.

SEC-CV · Control Valves · Diagnostics

CV diagnosis distinguishes demand generation, positioner response, actuator movement, and process effect. Agreement between demand and indicated travel does not establish that the closure element moves correctly or that the expected flow occurs.

Stick-slip behavior often appears as periods of unchanged travel followed by abrupt movement. Packing friction, mechanical binding, or linkage damage can contribute. Controller tuning changes do not remove the mechanical cause.

Slow response can follow inadequate air delivery, restricted exhaust, excessive actuator volume, or configured rate limitation. Compare supply pressure during movement with static pressure under the approved test plan.

Travel without the expected process effect can indicate damaged trim, coupling failure, changed pressure conditions, or an incorrectly sized restriction. Leakage at the closed position requires its own evidence and acceptance method.

Use recorded position, pressure, and process trends before invasive inspection. Any intervention controls stored energy and unexpected motion. The final record identifies the mechanism and verifies the relevant travel and functional requirements.

SEC-SV · Solenoid Valves · Definition

The canonical abbreviation is SV. The device uses an energized coil to create magnetic force that changes a fluid closure mechanism. Direct-acting construction moves the main closure directly; pilot-operated construction uses a controlled pilot passage and process pressure to move the main element.

The topic addresses discrete industrial switching. Specialized proportional constructions exist, so coil actuation itself does not imply a universal two-position limitation.

Normally open and normally closed describe the unenergized state under specified conditions. They do not independently establish a complete system's safe failure state.

Port arrangement determines whether the assembly isolates flow or redirects an actuator supply. Electrical coil rating, pressure differential, and fluid compatibility determine suitability together.

This device must not be confused with Control Valves (section SEC-CV), which provide commanded modulation of a flow passage as a final control function.

SEC-SV · Solenoid Valves · Principle

SV coil current establishes a magnetic field that attracts a movable armature. Magnetic force opposes spring force, fluid force, and friction. The resulting movement opens or closes the designated passage.

A direct-acting assembly must provide sufficient magnetic force across its allowed pressure differential. A pilot-operated assembly instead changes pressure distribution across a diaphragm or piston and may require a minimum differential.

A 24 V direct-current coil rated at 6 W draws approximately 0.25 A at its nominal steady condition. Temperature-dependent resistance and electronic drive can alter the current; alternating-current coils can also have substantially different inrush behavior.

Coil suppression reduces switching transients but may delay release. The chosen suppression method therefore affects the system's required response and must match the approved electrical design.

Audible movement confirms neither correct port routing nor leak-tight closure. The complete function requires the intended electrical response, mechanical motion, and fluid-path state under the specified pressure conditions.

SEC-SV · Solenoid Valves · Specification

The specification identifies port function, de-energized state, actuation method, pressure differential, and coil rating. The example describes a small direct-acting pneumatic switching assembly.

Table SEC-SV-1: Example discrete pneumatic duty.

Property	Example value
Nominal coil supply	24 V direct current
Rated steady coil power	6 W
Selected pressure differential	0 to 700 kPa
Illustrative switching time	30 ms under stated test conditions
Command mode; Control Valves	Two states at 0 V and 24 V

The comparison row describes this discrete example, not every available electromagnetic construction. Switching time requires pressure, fluid, voltage, and suppression conditions.

Specify seal material, contamination tolerance, duty cycle, enclosure protection, and hazardous-area approval where applicable. A compatible body material does not establish elastomer compatibility.

Minimum operating differential is especially important for pilot-operated designs. Output-driver capacity includes steady current, transient behavior, and permitted suppression. A nominal voltage match alone does not establish electrical suitability.

SEC-SV · Solenoid Valves · Installation

SV installation follows the designated port connections, flow direction, and mounting orientation. An incorrect port schedule can produce a functioning coil with the wrong actuator or process response.

Fluid cleanliness and filtration meet the assembly requirements. Sealants remain outside internal passages, and pipe loads do not distort the body. Exhaust outlets remain unobstructed and route as required by the application.

Coil voltage, frequency where applicable, and electrical polarity match the nameplate and suppression circuit. Protective bonding and cable sealing follow the approved electrical and environmental design.

A manual override can change the fluid state independently of the electrical command. Its presence, locking policy, and normal position require documentation. Override use never bypasses required authorization.

Commissioning proves energized and de-energized fluid functions under a safe test plan. Stored pressure and unexpected actuator movement remain controlled. Coil indication alone is insufficient acceptance evidence.

SEC-SV · Solenoid Valves · Calibration

SV evaluation normally establishes switching performance rather than a continuous measurement calibration. The test record distinguishes calibrated reference measurements of voltage, pressure, leakage, or time from verification of the assembly's discrete fluid function.

An approved isolated test fixture supplies compatible clean fluid within the rated pressure differential. The test plan identifies port allocation, coil rating, suppression arrangement, ambient temperature, and permissible cycling. No operating process protection is defeated for this test.

Record the as-found result at the specified energized and de-energized conditions. For the existing 24 V example, measure voltage at the coil terminals during operation rather than assuming the power supply setting reaches the load unchanged.

Where switching time is an acceptance requirement, define its endpoints. Electrical command onset, armature movement, downstream pressure response, and actuator movement are different events. A calibrated timing reference must observe the event named in the requirement.

Leakage verification uses a documented pressure differential, test medium, stabilization period, and measurement method. Audible operation does not establish closure integrity. A pilot-operated assembly also requires verification at its specified minimum operating differential.

Adjustment is limited to features explicitly provided for that purpose. Preserve initial results before repair or replacement, and document changed seals, coils, or suppression components. As-left verification repeats the relevant fluid states and acceptance conditions; the record retains reference identification and measurement uncertainty for reported quantities.

SEC-SV · Solenoid Valves · Diagnostics

SV diagnosis separates command availability, coil behavior, mechanical movement, and fluid routing. An illuminated connector indicates its own electrical state; it does not establish sufficient coil voltage or the intended pressure response.

No movement with an active command can follow an open winding, undervoltage, armature binding, or pressure differential outside the allowable envelope. Reference electrical observations to the actual terminal conditions through an authorized test arrangement.

Movement without downstream response suggests incorrect port connections, restricted passages, inadequate pilot differential, or a blocked exhaust. A click is evidence of a mechanical event, not proof that the main closure completes its travel.

Failure to release can involve retained electrical drive, suppression-induced delay, contamination, or a manual override left engaged. Document the override state before changing it. Unexpected movement remains a hazard after removal of the normal command.

Excessive coil temperature requires comparison with the rated duty and ambient conditions. A continuously energized coil can normally feel hot; touch is neither a safe temperature measurement nor a reliable acceptance criterion. Incorrect alternating-current operation can create additional heating.

Seat leakage may originate from particles, damaged seals, or incorrect differential direction. Inspection requires verified electrical and pressure isolation. The diagnostic record identifies the observed mechanism and the repaired component, then confirms the required energized and de-energized fluid states under the original acceptance conditions.

SEC-GWR · Guided Wave Radar Level Transmitters · Definition

The canonical abbreviation is GWR. The device determines surface position from the travel time of electromagnetic energy guided along a probe. A reflection occurs where electrical properties change, commonly at the boundary between vapour and liquid. Electronics convert the identified reflection delay into distance from a defined reference plane.

The reported level follows the configured relationship between measured distance and vessel datum. Probe length, vessel height, and zero-level reference are separate geometric quantities. Confusing them can create an apparently consistent but displaced indication.

Rod, cable, and coaxial probe constructions provide different mechanical and electromagnetic arrangements. Suitability depends on vessel geometry, dielectric properties, buildup, agitation, and pressure-boundary requirements rather than level span alone.

The principal measurand is distance to a detectable reflecting boundary. Interface measurement requires an appropriate upper-liquid transmission path, sufficient dielectric contrast, and an instrument configuration qualified for that service.

Empty-vessel reflections and near-probe obstructions can compete with the desired surface reflection. A valid signal requires correct identification, not merely detection of a strong echo. Configured inaccessible zones constrain the usable measurement interval.

This device must not be confused with Capacitive Level Switches (section SEC-CAP), which indicate a threshold condition from a capacitance change rather than continuously locating a surface by reflection travel time.

SEC-GWR · Guided Wave Radar Level Transmitters · Principle

GWR operation relates round-trip propagation time to distance along the guided path. For a known propagation velocity, distance equals one-half of that velocity multiplied by the measured delay. The factor of one-half accounts for outward and returning travel.

In a vapour space approximated by vacuum propagation, a surface 3 m below the reference plane gives a round-trip delay of approximately 20 ns. Actual processing also accounts for the instrument's internal reference and qualified environmental effects.

A dielectric discontinuity changes the electromagnetic impedance and returns part of the transmitted energy. Reflection strength depends on dielectric contrast and probe construction. A weak reflection is not equivalent to an absent surface.

The probe guides energy and limits beam spreading, but nearby conductive structures can still alter the response. Contacts, deposits, and changes in probe geometry introduce additional impedance discontinuities.

Interface measurements require energy to pass through the upper liquid before reaching the lower boundary. Propagation velocity within that upper liquid differs from the vapour-space value. Its dielectric properties therefore enter the distance calculation.

Echo selection, reference-plane geometry, and conversion to vessel level form distinct processing stages. A correct delay measurement can yield an incorrect level when the vessel datum is wrong. Damping smooths the reported value but cannot repair an incorrectly selected reflection.

SEC-GWR · Guided Wave Radar Level Transmitters · Specification

GWR selection identifies probe construction, dielectric conditions, vessel geometry, and the required reference plane. The example represents a moderate-temperature liquid service with a qualified rod probe; its values are application requirements, not universal technology ratings.

Table SEC-GWR-1: Illustrative guided-distance application.

Property	Example value
Usable measured distance	0.3 to 4.0 m
Reference-condition error target	±3 mm
Selected liquid temperature	−10 to 120 °C
Example pressure requirement	1.6 MPa
Output; SEC-CAP distinction	Continuous 0.3 to 4.0 m

The comparison property describes continuous distance reporting rather than a threshold state. The section identifier avoids treating the two measurement functions as interchangeable.

State the minimum dielectric conditions and required echo margin for the actual medium. Upper and lower inaccessible zones depend on probe construction and operating conditions; their dimensions must come from the selected assembly.

Probe and seal materials require process compatibility. Stainless steel can suit many clean liquids, but it establishes neither universal corrosion resistance nor suitability for sticky deposits.

Mechanical loading, cable restraint where applicable, and chamber geometry require separate limits. Pressure rating belongs to the complete process connection and seal assembly. A favourable distance-error specification does not establish reliable detection through foam, buildup, or an unsuitable interface.

SEC-GWR · Guided Wave Radar Level Transmitters · Installation

GWR installation establishes a documented reference plane and a stable guided path. The approved vessel drawing identifies nozzle geometry, probe clearance, usable length, and the level datum. Vessel entry and pressure-boundary work require authorized isolation and access controls.

The probe avoids unintended contact with walls, internal supports, and moving equipment. Rod bending or cable movement can create unwanted reflections and mechanical damage. Restraint, where required, follows the qualified design rather than an improvised attachment.

Nozzle dimensions and nearby obstructions affect the upper measurement region. An external chamber must communicate representative liquid conditions and avoid an unrecognized temperature or density difference from the vessel.

Confirm process connection and seal compatibility before installation. Probe shortening or replacement changes geometry and may require revised configuration or qualification. Neither action is an ordinary zero adjustment.

Commissioning records probe construction, reference elevation, inaccessible zones, medium properties, and echo-selection settings. Capture representative echo evidence under known conditions. False-echo suppression must not mask the valid surface interval or conceal a mechanical contact.

The installation record links the measured distance to the vessel datum and receiving-system scaling. Verification uses an independent level reference where safely available; assumptions about an empty vessel require recorded evidence.

[FIGURE SLOT: vessel reference plane, guided probe, clearances, and level datum]

SEC-GWR · Guided Wave Radar Level Transmitters · Calibration

GWR calibration distinguishes measured distance, derived level, and transmitted output. Each stage needs a defined reference. A current-output comparison alone does not establish surface-distance performance.

A suitable test arrangement compares indicated distance with independently established surface positions while preserving the qualified probe geometry. The reference record identifies the measurement plane, surface stability, and uncertainty in the physical distance.

For the 0.3 to 4.0 m example, selected distances of 0.5, 2.0, and 3.8 m cover separated portions of the usable interval. These points avoid assuming that an inaccessible zone provides a valid calibration position.

Vessel verification uses an independent level reference obtained without unsafe access or opening a live pressure boundary. Convert that reference to the same datum before calculating error. Thermal expansion and surface motion can limit comparison quality.

Electronic simulation evaluates reporting and output stages only. An echo-analysis check supports signal interpretation but is not itself a metrologically traceable physical level comparison. The record states the boundary of the evaluation.

Retain as-found geometry, echo settings, and results before adjustment. A datum correction addresses a documented geometric error, not an unexplained reflection shift. As-left verification confirms distance, level conversion, and receiving-channel endpoints over the relevant interval, with the reference identification and measurement uncertainty retained.

SEC-GWR · Guided Wave Radar Level Transmitters · Diagnostics

GWR diagnosis examines echo evidence before changing the reported level offset. A stationary indication can reflect selection of a nozzle or probe-end reflection rather than the moving liquid surface.

Lost surface detection can follow reduced dielectric contrast, foam, deposits, probe damage, or operation within an inaccessible zone. A healthy electronic self-check does not eliminate these physical explanations.

A sudden apparent level step may arise when echo selection transfers between competing reflections. Correlate the event with surface position, coating history, and configuration changes. Increased damping may hide the transfer without restoring correct identification.

An approximately constant distance error directs attention to the reference plane or probe configuration. An error that grows through an upper liquid layer instead suggests an unsuitable propagation model for interface service.

Probe contact with vessel internals can produce intermittent reflections or an enduring false boundary. Inspection requires safe isolation and authorized access. Cleaning methods preserve the probe, seals, and process compatibility.

Restore configuration from documented evidence rather than repeated trial adjustment. The final diagnostic record identifies the selected surface reflection, geometry, and observed fault mechanism. Verify recovery against a known surface condition and confirm that alarm behavior remains appropriate when valid detection is unavailable.

SEC-CAP · Capacitive Level Switches · Definition

The canonical abbreviation is CAP. The device detects a level threshold from a change in capacitance associated with material surrounding a sensing electrode. The electrode and its electrical reference form a capacitor whose response depends on geometry and the medium's electrical properties. Electronics compare that response with a configured switching criterion.

The reported state indicates whether the sensing condition crosses the threshold. It does not inherently provide continuous vessel level. The physical switching elevation depends on electrode geometry, insertion, medium properties, and the selected criterion.

Conductive liquids commonly require an insulated electrode construction. Nonconductive liquids and solids require sufficient dielectric contrast and a suitable reference arrangement. The same electrode is not automatically suitable for every medium.

Coating can remain on the sensing surface after the bulk material falls. That coating may change capacitance enough to influence the state. A design qualified for buildup compensation still has application limits.

Switching hysteresis prevents repeated state changes close to the threshold. Delay adds a timing criterion but does not change the underlying sensing principle. De-energized output behavior and alarm direction require explicit configuration.

This device must not be confused with Guided Wave Radar Level Transmitters (section SEC-GWR), which continuously locate a reflecting boundary from electromagnetic travel time along a probe.

SEC-CAP · Capacitive Level Switches · Principle

CAP sensing follows the dependence of capacitance on electrode geometry and permittivity. In simplified parallel geometry, capacitance is proportional to permittivity and area and inversely proportional to separation. Industrial probes use different geometry, but the same physical dependencies remain relevant.

Replacing air around an electrode with a liquid of greater relative permittivity commonly increases capacitance. The magnitude depends on the actual electrical field distribution, not solely on the volume of material in the vessel.

For conductive service with an insulated electrode, the insulation and surrounding conductive liquid establish the effective capacitor. Damage to that insulation can change the sensing behavior and create an unintended conductive path.

The sensing circuit may evaluate a frequency, charge, or impedance response related to capacitance. A threshold converts that continuous electrical response into a discrete state. Internal processing does not make the state an independent measure of all vessel contents.

Hysteresis establishes different transition criteria for rising and falling response. A delay requires the condition to persist before reporting a transition. These functions address chatter and transient exposure rather than contamination or poor reference geometry.

Changes in composition, moisture, temperature, or deposits can alter the response at unchanged level. A reliable threshold therefore rests on characterized covered and uncovered conditions for the actual medium, including its credible variations.

SEC-CAP · Capacitive Level Switches · Specification

CAP selection establishes covered and uncovered response, electrode construction, and required output state. The example describes a threshold application in a clean liquid; electrical values illustrate a selected configuration rather than a universal sensing envelope.

Table SEC-CAP-1: Illustrative liquid-threshold configuration.

Property	Selected example
Illustrative sensing increment	10 pF
Electrode insertion length	150 mm
Configured transition delay	0.5 s
Selected medium temperature	0 to 80 °C
Function; SEC-GWR distinction	Threshold at 1.2 m datum

The comparison row describes a state associated with one configured elevation, not continuous surface distance. The elevation is a geometric application value and does not imply millimetre-level switching repeatability.

Specify pressure capability, insulation material, process reference, and expected dielectric variation. Polytetrafluoroethylene insulation may suit selected liquids when its mechanical and chemical limits are met.

Output contacts require voltage, current, load type, and isolation ratings. A nominal relay current is insufficient for selecting an inductive load. Alarm direction and response to loss of supply require explicit acceptance criteria.

Required repeatability follows tests with the actual medium and mounting. Sensitivity, hysteresis, and buildup tolerance remain distinct specifications. A capacitance increment alone cannot establish whether the material will reliably generate that increment.

SEC-CAP · Capacitive Level Switches · Installation

CAP installation locates the sensitive region at the required threshold and establishes the intended electrical reference. The approved drawing identifies electrode orientation, insertion, vessel construction, and the relationship to the alarm datum.

A metallic vessel may form the electrical reference where the design permits. An insulating vessel can require a separate reference electrode. Protective earthing alone does not establish the intended sensing geometry.

Keep the sensitive region outside persistent inlet jets and deposits unless the assembly is qualified for that exposure. Material bridging between the electrode and a nearby structure can create a covered indication without the intended bulk level.

Process threads, seals, and insulation require compatible handling and tightening. Damage to the insulated sensing surface can invalidate conductive-liquid operation. Installation does not include exposing the electrode to an unapproved insulation test.

Commissioning records uncovered and covered responses with representative material. Confirm threshold, hysteresis, transition delay, and output state against the approved functional requirement. Tests that change a process alarm require authorization and restoration evidence.

Retain the physical mounting dimensions and reference arrangement with the configuration record. A later change in vessel lining, electrode length, or material composition can require renewed qualification even if the output still changes during a simple check.

[FIGURE SLOT: threshold elevation, insulated electrode, and vessel reference path]

SEC-CAP · Capacitive Level Switches · Calibration

CAP evaluation establishes the electrical threshold and verifies its relationship to the material level. A calibration of the electrical input and a functional check with the actual medium address different boundaries of the system.

Where the instrument supports a capacitance reference, compare the transition with a calibrated reference through the approved test interface. Parasitic capacitance in leads and fixtures belongs in the reference arrangement and measurement uncertainty.

Material-based verification observes rising and falling transitions relative to an independently established level. Record composition, temperature, mounting geometry, and surface behavior. A test with water does not automatically qualify an electrode intended for a different liquid.

For a threshold selected at 1.2 m, report actual transition elevations rather than merely stating that the switch operates. The difference between rising and falling elevations includes sensing hysteresis and physical wetting effects under those conditions.

Verify configured delay with a suitable timing reference when timing matters. Define whether the interval starts at physical coverage, electrical criterion crossing, or an external command; those starting events are not interchangeable.

Preserve the as-found result before changing sensitivity or threshold. Adjustment must retain separation between credible uncovered and covered responses. As-left verification includes the alarm direction and supply-loss state, while the record identifies reference capability and the limits of extrapolation to other material conditions.

SEC-CAP · Capacitive Level Switches · Diagnostics

CAP diagnosis compares the physical material condition with the electrical sensing response and reported state. A persistent covered indication can reflect residual coating rather than a high bulk level.

An uncovered indication during genuine coverage can follow inadequate dielectric contrast, a missing reference path, damaged insulation, or excessive threshold setting. Confirm the medium and mounting before assigning the fault to electronics.

Repeated transitions near the threshold may arise from waves, intermittent bridging, or insufficient hysteresis. Delay can reduce chatter but must remain consistent with the required alarm response. Increasing delay without review can weaken the functional requirement.

An abrupt response change after maintenance directs attention to electrode length, cable routing, vessel reference, or insulation damage. A configuration copied from another installation may not preserve the same electrical geometry.

Separate sensing status from contact interpretation at the receiving system. Incorrect normally-open or normally-closed allocation can create an inverted alarm despite valid detection. Supply-loss behavior also requires comparison with the documented design.

Cleaning and inspection occur only after approved process and electrical isolation. Avoid trial sensitivity changes that hide deposits or loss of reference. Record the diagnosed mechanism and verify restored covered, uncovered, and failure states with representative material and the original mounting arrangement.

SEC-PH · pH Sensors · Definition

The canonical designation is pH. It denotes the dimensionless quantity defined by the negative base-ten logarithm of hydrogen-ion activity. The device addressed here uses a hydrogen-ion-responsive glass membrane and a reference electrode to establish an electrical potential related to that quantity in an aqueous medium.

The measuring electrode and reference electrode form an electrochemical system. A combined assembly contains both functions within one body, but their roles remain distinct. The reference junction provides ionic contact with the sample.

The electrical signal requires a high-input-impedance measurement circuit. Cable leakage, connector contamination, and inadequate shielding can disturb the result even when the glass membrane remains intact.

Temperature affects electrode sensitivity and can also change the sample's own pH. Electrode temperature compensation addresses the first effect; it does not generally convert every sample result to its value at another temperature.

The result describes the sampled liquid under the stated conditions. Dilution, gas exchange, pressure reduction, or cooling during sampling can change those conditions. A laboratory comparison is meaningful only when sample alteration is controlled or acknowledged.

This device must not be confused with ORP (Redox) Sensors (section SEC-ORP), which report an electrode potential associated with oxidation and reduction conditions rather than hydrogen-ion activity.

SEC-PH · pH Sensors · Principle

pH measurement relates electrode potential to hydrogen-ion activity through an approximately Nernstian response. For an ideal monovalent response, the sensitivity magnitude is about 59.16 mV per pH unit at 25 °C. Signal polarity follows the electrode and measuring-circuit convention.

The response depends on absolute temperature. At higher temperature the ideal sensitivity magnitude increases. The associated temperature measurement therefore enters electrode conversion, while error in that temperature can contribute to the reported pH.

The glass membrane develops its selective response through hydrated surface layers. Drying, chemical attack, ageing, and contamination can alter sensitivity, response time, or electrical resistance. A physically unbroken membrane can still perform poorly.

The reference electrode supplies a comparatively stable potential and ionic connection. Junction contamination or sample-dependent liquid-junction potential can influence the total signal independently of the glass response.

Because the membrane circuit can have very high resistance, minute leakage currents matter. Moist connectors or damaged cable insulation can create measurement error without an obvious short circuit.

Two reference buffers establish an observed offset and slope over their interval. They do not eliminate every matrix effect in the process liquid. Extreme alkalinity, low ionic strength, or a significant difference between buffer and sample conditions can introduce limitations requiring application-specific evaluation.

SEC-PH · pH Sensors · Specification

pH selection specifies sample chemistry, temperature, pressure, reference construction, and measuring-circuit compatibility. The selected interval below represents a moderate aqueous process, not the unrestricted capability of every glass assembly. pH values remain dimensionless; units apply to electrical and environmental quantities.

Table SEC-PH-1: Illustrative aqueous measurement conditions.

Property	Illustrative value
Selected pH interval	2 to 12, dimensionless
Ideal sensitivity at 25 °C	59.16 mV per pH unit
Selected sample temperature	5 to 70 °C
Example glass resistance at 25 °C	100 MΩ
Response; SEC-ORP distinction	H-ion signal in mV

The comparison property identifies the selective electrochemical response, not a generic potential measurement. The ideal sensitivity is a physical reference value rather than an individual electrode's guaranteed slope.

Specify the permissible observed slope and offset through the application acceptance procedure. Buffer uncertainty, temperature measurement, junction behavior, and sample handling contribute to complete measurement uncertainty.

Body material, reference electrolyte, glass formulation, and process seals require chemical compatibility. A temperature rating alone does not establish resistance to cleaning chemistry or pressure cycling.

Required response time includes its test medium, temperature, and stabilization criterion. Low-conductivity service and abrasive or coating liquids may require specialized construction. Electrical input capability must accommodate the actual membrane resistance and leakage limits.

SEC-PH · pH Sensors · Installation

pH installation presents representative liquid to the membrane and reference junction while maintaining the approved pressure boundary. The mounting arrangement allows safe removal, cleaning, and reference verification without relying on unsafe access to a live process.

Orientation follows the assembly instructions so that internal electrolyte and hydrated glass remain in their intended condition. Immersion covers the active membrane and reference junction. Gas pockets or intermittent exposure can interrupt representative measurement.

A flow-through assembly limits uncontrolled sampling changes and provides suitable liquid exchange. Sample cooling, pressure reduction, or gas release may change the measured quantity; the design records which conditions the result represents.

Keep high-impedance connectors clean and dry. Use the qualified sensor cable and termination arrangement, avoiding uncontrolled splices. Grounding and isolation follow the documented design to limit unwanted electrical paths through the process.

The temperature element occupies a representative thermal region. Its wiring and configuration match the receiving circuit. Temperature compensation does not justify ignoring a temperature difference between the electrode and sample.

Commissioning records initial buffer results, slope, offset, sample conditions, and installation geometry. Protective devices and extraction assemblies follow their authorized operating procedures. Storage and cleaning preserve membrane hydration and reference integrity using the approved fluids rather than an assumed universal method.

[FIGURE SLOT: representative liquid path, glass membrane, reference junction, and temperature element]

SEC-PH · pH Sensors · Calibration

pH calibration establishes electrode offset and observed slope against reference buffers at measured temperature. Buffer certificates provide assigned values, temperature dependence, expiry, and measurement uncertainty. Nominal bottle labels alone do not establish the reference value under the test conditions.

Use fresh aliquots and a clean, hydrated assembly under an approved isolated procedure. Rinsing prevents transfer between buffers; handling avoids abrasion of the membrane. Returned liquid does not re-enter the stock container.

An illustrative pair has assigned values near 7.00 and 4.01 at 25 °C. Select references that appropriately bracket the intended sample interval. Record actual certificate values rather than imposing these nominal examples on every buffer formulation.

Observe stable raw potential and temperature at each point. Express the observed slope in mV per pH unit and compare its magnitude with the temperature-dependent ideal response. Acceptance limits follow the documented application procedure, not an assumed universal percentage.

Preserve the as-found result before adjustment. Where cleaning is necessary, retain results both before and after cleaning so that contamination effects remain distinguishable from electronic correction. A simulator verifies the receiving circuit but excludes membrane and reference-junction performance.

As-left verification uses an independent reference condition where practicable. Report reference identification, stabilization criterion, slope, offset, and measurement uncertainty. A successful buffer comparison does not remove sample-specific junction effects or changes caused by sampling.

SEC-PH · pH Sensors · Diagnostics

pH diagnosis distinguishes membrane response, reference behavior, electrical leakage, and representative sampling. Disagreement with a laboratory result requires examination of sample temperature, gas exchange, and elapsed time before either channel is judged defective.

A declining observed slope can accompany membrane ageing, dehydration, or surface contamination. Slow stabilization can arise from deposits or poor liquid exchange. The same symptom can have more than one mechanism, so retain raw potential and temperature observations.

A large offset with otherwise credible slope directs attention to the reference system and connector leakage. A blocked junction may introduce sample-dependent error even when buffer readings appear satisfactory.

Erratic signals can correlate with wet connectors, cable movement, electrical interference, or intermittent immersion. High-impedance circuits are sensitive to leakage that an ordinary continuity check may not reveal.

An abrupt change after maintenance warrants review of temperature configuration, cable allocation, reference connection, and stored calibration values. A software offset must not conceal damaged glass or an unstable reference.

Approved cleaning and isolated comparison separate recoverable contamination from irreversible deterioration. Replacement preserves assembly identification and as-found evidence. The diagnostic record confirms restored reference response and representative process indication without claiming that buffer agreement eliminates every sample-matrix effect.

SEC-ORP · ORP (Redox) Sensors · Definition

The canonical abbreviation is ORP. The device measures the electrical potential of an electrochemically active measuring electrode relative to a specified reference electrode. The observed potential reflects interactions with oxidizing and reducing species at the electrode surface under the sample conditions.

The measurand is electrode potential in volts, commonly reported in millivolts. It is not a universal concentration measurement. Converting potential into concentration requires a justified chemical model and controlled conditions that many industrial mixtures do not satisfy.

Platinum and gold are possible measuring-electrode materials. Their surface behavior can differ with sample chemistry. The reference construction and electrolyte also affect the reported potential, so results require an identified reference basis.

A mixed sample can establish a mixed potential rather than equilibrium with one isolated chemical couple. Surface contamination, reaction kinetics, and dissolved gases influence the result. A stable display therefore does not independently establish thermodynamic equilibrium.

Temperature changes reaction behavior and reference potential as well as any applicable equilibrium relationship. There is no universal temperature correction that converts every industrial sample to a common reference temperature.

This device must not be confused with pH Sensors (section SEC-PH), which selectively respond to hydrogen-ion activity through a glass-membrane electrochemical system.

SEC-ORP · ORP (Redox) Sensors · Principle

ORP arises from charge-transfer interactions at the measuring-electrode surface. For a defined reversible chemical couple at equilibrium, the Nernst relationship connects potential with temperature, electron number, and the activity ratio of oxidized and reduced forms.

At 25 °C, a tenfold change in that ratio changes the equilibrium potential by approximately 59.16 mV divided by the electron number, with sign determined by the stated reaction convention. This relationship does not authorize concentration conversion for an unidentified mixture.

The measuring circuit observes the difference between the measuring electrode and reference electrode. A reported value relative to a silver–silver-chloride reference is not numerically identical to a value relative to the standard hydrogen electrode.

Reference conversion requires the relevant reference-electrode potential, including electrolyte and temperature conditions. A fixed correction copied from an unspecified reference can introduce a systematic error.

In a mixed sample, several reactions can contribute different exchange rates. The resulting potential depends on surface condition and kinetics as well as composition. Agitation and oxygen exchange can change the observed response.

The instrument's input circuit should draw negligible current from the electrochemical system. Stable electronics cannot correct electrode fouling or establish chemical equilibrium. Interpretation therefore retains the reference basis, temperature, and measurement conditions alongside the numerical potential.

SEC-ORP · ORP (Redox) Sensors · Specification

ORP selection identifies measuring metal, reference construction, electrolyte, sample composition, and output basis. The configuration below describes a moderate aqueous application. Its limits represent training examples rather than simultaneous guarantees for every electrochemical assembly.

Table SEC-ORP-1: Representative potential-measurement channel.

Property	Example value
Selected potential interval	−1000 to +1000 mV
Electronics error target	±1 mV
Selected sample-temperature interval	10 to 60 °C
Example pressure requirement	600 kPa
Measurand; pH Sensors	Electrode potential in mV

The comparison row identifies potential reporting rather than a hydrogen-ion activity result. The electronics error target excludes reference instability, junction effects, surface kinetics, and sample alteration. These contributions can dominate the complete measurement uncertainty.

Specify platinum or another qualified measuring material from the actual chemistry. A corrosion-resistant body does not establish resistance of the electrode surface, reference junction, or seals to poisoning and cleaning agents.

The reference description includes electrolyte concentration and construction where those details determine the potential basis. A receiving system label must not imply conversion to another reference unless that conversion is explicitly implemented and justified.

Response-time requirements identify test solution, temperature, mixing, and stabilization criterion. Alarm or control setpoints depend on qualified process evidence; a generic millivolt value cannot establish chemical treatment effectiveness in every sample.

SEC-ORP · ORP (Redox) Sensors · Installation

ORP installation exposes the measuring surface and reference junction to representative liquid. The approved arrangement maintains immersion and adequate replacement of liquid around the assembly without excessive erosion or mechanical loading.

Mounting orientation follows reference-electrolyte and junction requirements. A gas pocket can interrupt ionic contact or change local chemistry. The pressure-containing holder and extraction system require their own qualified operating procedure.

Sample conditioning can alter the potential through cooling, pressure reduction, oxygen exchange, or reaction during transport. Record the conditions represented by the measurement. A conditioned sample is not automatically equivalent to the vessel liquid.

Use the qualified cable and keep connectors clean and dry. Grounding and electrical isolation follow the documented process arrangement. Unintended current through the measuring surface can alter electrochemical behavior.

Cleaning access permits safe removal without opening a live boundary. Cleaning materials and methods match the measuring metal, reference junction, and seals. Abrasive treatment is not assumed acceptable.

Commissioning records the reference basis, initial solution check, temperature, response criterion, and physical location. Confirm signal interpretation at the receiver. Protective-function testing requires authorization, and restoration includes the documented configuration and process boundary.

[FIGURE SLOT: liquid flow path, measuring metal, reference junction, and isolated holder]

SEC-ORP · ORP (Redox) Sensors · Calibration

ORP evaluation separates electrical potential calibration from verification of the electrochemical assembly. A calibrated millivolt source assesses the receiving circuit; it does not establish the behavior of the measuring surface or reference junction.

A suitable reference solution has an assigned potential for stated temperature and reference-electrode conditions. Use the certificate's value and uncertainty. Different reference constructions can require different expected values for the same solution.

An illustrative electrical sequence applies −500, 0, and +500 mV through the approved isolated input arrangement. Record indicated potential, output response, polarity, and input configuration. The source must be compatible with the circuit under test.

For an assembly check, use a fresh uncontaminated solution aliquot and record stabilization behavior. Surface cleaning follows the qualified method, and initial observations are retained before cleaning when as-found evidence is required.

A one-solution check can reveal an offset or poor response but does not establish a universal process slope. Applying an arbitrary correction may hide a deteriorated reference or a contaminated measuring metal.

As-left verification retains the reference basis, solution certificate, temperature, and acceptance criterion. Report measurement uncertainty for the defined comparison and distinguish it from unknown sample-matrix effects. No general agreement in a check solution proves concentration accuracy or equilibrium in the operating process.

SEC-ORP · ORP (Redox) Sensors · Diagnostics

ORP diagnosis examines surface condition, reference behavior, sample chemistry, and electrical reporting. A process potential change can be genuine even when another chemical measurement remains constant.

Slow response may follow deposits, surface poisoning, inadequate sample exchange, or intrinsically slow electrochemical reactions. Increasing signal damping cannot distinguish these mechanisms and can conceal their timing.

A persistent offset between two assemblies requires comparison of reference bases before fault assignment. Different electrolyte conditions or junction behavior can produce different readings in otherwise similar arrangements.

Erratic potential can correlate with gas contact, grounding disturbance, connector moisture, or intermittent reference contact. Retain raw millivolt and temperature trends with operating events rather than viewing the processed value alone.

Recovery in a reference solution but disagreement in the process suggests sample-dependent behavior or sampling differences. It does not automatically establish that the process value should be corrected to match the solution check.

Approved isolated cleaning and reference comparisons support a documented diagnosis. Avoid unqualified offsets that mask electrode deterioration. The record identifies changed components, retained reference basis, and restored response under known conditions, while preserving uncertainty about uncharacterized chemical mixtures.

SEC-PIEZO · Piezoelectric Vibration Sensors · Definition

The canonical abbreviation is PIEZO. The device converts dynamic mechanical loading of a piezoelectric element into electrical charge. In the accelerometer construction addressed here, an internal seismic mass applies force to the element as the mounting base accelerates. The calibrated output represents acceleration along a specified sensitive axis.

The measurand is acceleration at the mounting location within a qualified dynamic interval. It is not inherently shaft-relative displacement or a direct measure of every machine condition.

Charge-output assemblies require a suitable charge amplifier. Assemblies containing electronics provide a conditioned voltage and require a specified excitation arrangement. These constructions have different temperature and cable constraints.

Piezoelectric conversion does not provide sustained static-acceleration measurement. Electrical leakage and conditioning establish the low-frequency limit, while mechanical resonance and mounting establish important upper-frequency limitations.

The mounting interface transfers motion from the structure to the sensor. Surface preparation, fastening, orientation, and cable restraint therefore belong to the measurement system. A calibrated sensor on an unsuitable mounting can produce misleading spectra.

This device must not be confused with Eddy Current Proximity Probes (section SEC-EDDY), which measure the gap to a conductive target without mechanical contact.

SEC-PIEZO · Piezoelectric Vibration Sensors · Principle

PIEZO accelerometer operation uses force generated by the seismic mass during acceleration. Over the qualified interval, the element's charge is approximately proportional to that force. Mechanical design controls preload, resonance, transverse response, and susceptibility to base strain.

A charge amplifier converts charge to voltage through a defined feedback arrangement. Sensor and cable capacitance influence practical noise and stability even where the intended charge conversion minimizes sensitivity to capacitance changes.

Integrated electronics condition the high-impedance element signal near its source. A shared cable can carry excitation and the dynamic signal on a bias voltage. The receiver must provide compatible current and sufficient voltage compliance.

An illustrative sensitivity of 10.2 mV/(m/s^2) produces about 51 mV peak for 5 m/s^2 peak acceleration. RMS, peak, and peak-to-peak amplitudes describe different quantities and require explicit labeling.

For sinusoidal motion, velocity sensitivity derived by integration depends on vibration frequency. Broadband integration also requires suitable filtering and cannot recover a static component absent from the original signal.

Response near mechanical resonance can greatly exceed the flat-response sensitivity. A loose mounting or flexible adapter introduces additional dynamics. Qualified frequency response therefore applies to the specified assembly and mounting conditions rather than every attachment method.

SEC-PIEZO · Piezoelectric Vibration Sensors · Specification

PIEZO selection defines measured axis, amplitude convention, frequency response, mounting, and conditioning. The example represents an industrial accelerometer with integrated electronics. Values are a plausible configuration and require confirmation against the actual calibrated assembly.

Table SEC-PIEZO-1: Illustrative dynamic-acceleration channel.

Property	Example requirement
Nominal acceleration sensitivity	10.2 mV/(m/s^2)
Selected frequency interval	10 to 5000 Hz
Example dynamic amplitude limit	±490 m/s^2 peak
Conditioning supply current	4 mA
Static response; SEC-EDDY	No sustained 0 Hz response

The comparison row identifies the absence of static response in this conversion method. The upper amplitude remains subject to frequency, mechanical limits, electronic headroom, and the receiver's input range. It is not a universal machinery acceptance threshold.

Specify sensitivity tolerance or calibration uncertainty separately from frequency-response deviation. Transverse sensitivity, mass loading, base-strain response, environmental sealing, and temperature influence also require explicit limits where relevant.

A stainless steel housing may suit industrial exposure, but connector protection and mounting corrosion can still control service reliability. Integrated electronics can impose a lower usable temperature limit than the piezoelectric element itself.

Cable and conditioner requirements include excitation compliance, bias interpretation, bandwidth, and overload recovery. Sampling frequency and anti-alias filtering belong to the receiving system. A wide sensor bandwidth does not establish correctly sampled spectral evidence.

SEC-PIEZO · Piezoelectric Vibration Sensors · Installation

PIEZO installation provides a rigid, reproducible interface at the documented measurement location. Mounting work occurs on safely isolated equipment, with access and stored-energy controls required by the machinery procedure.

The sensitive axis follows the intended measurement direction. Record axial, radial, or other geometry without assuming that one orientation represents all motion. Nearby structural resonances can influence the measured housing acceleration.

A prepared mounting pad and qualified fastening arrangement support the intended bandwidth. Tightening follows the assembly instructions. Excessive force can damage threads or the sensor base, while insufficient fastening can introduce resonance and intermittent contact.

Magnetic mounts and adhesive fixtures have their own frequency and environmental limitations. Their use must not inherit the bandwidth established for a rigid stud mounting without evidence.

Restrain cables to limit motion, connector stress, and interference from adjacent surfaces. Charge-output cable behavior differs from low-impedance conditioned output, so cable selection follows the actual construction.

Commissioning records location, orientation, fixture, sensitivity, excitation, and bias condition. Confirm receiving bandwidth and amplitude units before establishing a baseline. Operational observations remain outside moving-equipment exclusion zones and do not bypass machine protection.

[FIGURE SLOT: rigid mounting pad, sensitive axis, approved fastener, and cable restraint]

SEC-PIEZO · Piezoelectric Vibration Sensors · Calibration

PIEZO calibration compares electrical response with a known dynamic acceleration. A comparison facility uses a qualified reference sensor or other established motion reference and reports the mounting and excitation conditions.

An illustrative sensitivity comparison at 160 Hz and 10 m/s^2 RMS establishes response at one condition. It does not independently verify the complete 10 to 5000 Hz operating interval. Frequency-response evaluation requires additional justified points.

Mount the assembly using the qualified fixture and control transverse motion, distortion, and mechanical loading. The reference and test sensor must experience sufficiently equivalent motion for the stated comparison method.

Record conditioner gain, excitation, cable, bias, and receiver input settings. Report sensitivity with its acceleration and voltage amplitude conventions. A peak voltage divided by an RMS acceleration produces an inconsistent sensitivity.

A portable vibration source supports a field check within its qualified conditions. Electrical injection assesses downstream reporting but excludes the piezoelectric element and mechanical transmission path.

Retain the as-found result before any permitted adjustment of the receiving sensitivity. Most sensor elements are not field-adjustable. As-left verification confirms the complete channel conversion and documents reference identification, frequency, amplitude, environmental conditions, and measurement uncertainty. Metrological traceability applies to the defined calibration result, not every subsequent spectral diagnosis.

SEC-PIEZO · Piezoelectric Vibration Sensors · Diagnostics

PIEZO diagnosis separates sensor health, mounting transmission, signal conditioning, and actual structural motion. A changed spectrum alone does not identify which stage changes.

Abnormal bias in a conditioned assembly can indicate insufficient excitation, an open cable, a short, or failed electronics. Interpretation uses the specified bias limits and conditioner behavior rather than a universal voltage threshold.

A distorted waveform can reflect acceleration overload, inadequate voltage headroom, or receiver clipping. Large peaks near a resonance can also exceed the flat-response assumption. Check the entire gain chain before interpreting harmonic content.

Unexpected narrow spectral components may originate from a loose fastening, flexible fixture, or cable contact. Changes after remounting require comparison against the original location and mounting evidence.

Apparent high-frequency components can result from aliasing when acquisition filtering and sample rate are unsuitable. Sensor bandwidth does not protect a receiver from undersampling. Integrated velocity can also exaggerate low-frequency noise.

Diagnosis uses approved isolated inspections and reference checks without approaching moving equipment. Preserve settings and baseline evidence before replacement. The final record verifies restored bias, mounting, sensitivity, and acquisition conditions, and distinguishes instrument recovery from any unverified conclusion about machine condition.

SEC-EDDY · Eddy Current Proximity Probes · Definition

The canonical abbreviation is EDDY. The device determines separation from a conductive target through changes in the electrical behavior of an excited probe coil. Eddy currents induced in the target alter the coil's impedance, and a matched conditioning system converts that change into a gap-related signal.

The measurand is distance between the probe face and the qualified target surface. The output can represent static gap and dynamic variation within the specified linear and frequency intervals.

The probe, extension cable, and conditioner form a matched system. Their electrical characteristics and total cable length influence sensitivity. Components with visually compatible connectors are not necessarily interchangeable.

Target conductivity, magnetic properties, shape, and surface condition matter. A calibration obtained on one alloy does not automatically establish response on another material or a thin coating.

For shaft applications, the measurement is relative to the structure supporting the probe. Mechanical and electrical runout can contribute to the observed signal, so dynamic gap variation is not identical to vibration of the whole machine in an inertial frame.

This device must not be confused with Piezoelectric Vibration Sensors (section SEC-PIEZO), which derive dynamic acceleration from force applied to an internal piezoelectric element.

SEC-EDDY · Eddy Current Proximity Probes · Principle

EDDY excitation establishes a high-frequency magnetic field near the probe face. A conductive target supports circulating induced currents that oppose the exciting field. Their effect on coil impedance depends on separation and target properties.

The conditioner interprets the resulting electrical change through a characterized relationship. Only part of the complete response is sufficiently linear for the stated sensitivity. Operation outside that interval can distort dynamic amplitude and invalidate static gap interpretation.

An illustrative sensitivity magnitude of 7.87 V/mm gives about 0.787 V change for 0.10 mm displacement within the linear region. Voltage direction follows the specified conditioner convention and must not be inferred from magnitude alone.

Excitation penetrates a material-dependent depth. Conductivity, permeability, and excitation frequency influence that depth and the response to coatings or thin targets. Surface geometry also changes the electromagnetic interaction.

The static output defines the operating gap, while its alternating component represents dynamic gap variation. A suitable initial gap retains room for the expected movement without leaving the linear interval.

Electrical runout arises from variations in target properties, and mechanical runout arises from geometric variation. Both can appear during rotation. The observed waveform therefore combines true relative motion with target-dependent contributions that require separate evidence.

SEC-EDDY · Eddy Current Proximity Probes · Specification

EDDY selection treats probe, cable, conditioner, and target as a qualified combination. The example describes a steel-target system with a millimetre-scale linear gap interval. Its values are representative rather than interchangeable specifications for all probe diameters.

Table SEC-EDDY-1: Illustrative conductive-target system.

Property	Representative value
Qualified linear gap interval	0.5 to 2.5 mm
Sensitivity magnitude	7.87 V/mm
Selected dynamic bandwidth	0 to 8000 Hz
Matched total cable length	5 m
Static capability; SEC-PIEZO	Gap response at 0 Hz

The comparison row identifies a static-distance capability. Dynamic bandwidth does not extend the permitted gap interval, and the stated sensitivity requires the qualified target material and geometry. Output polarity is separately defined by the selected conditioner.

Specify probe-face diameter, target dimensions, mounting clearance, and permitted adjacent metal. Nearby conductive material can alter the field and generate cross-interference between probes. Material surrounding the probe is therefore part of installation qualification.

Temperature limits apply independently to probe, cable, connector, and conditioner. Environmental sealing and material compatibility require assessment for the actual lubricant, cleaning fluid, and enclosure conditions.

Linearity deviation and sensitivity uncertainty are distinct from machine runout. Metallurgical variation, coating thickness, and surface damage can affect target response. A narrow electronics error specification cannot establish total displacement uncertainty on an unqualified shaft surface.

SEC-EDDY · Eddy Current Proximity Probes · Installation

EDDY installation establishes a rigid probe support and qualified target geometry. Work near shafts occurs only under authorized machinery isolation, with stored-energy and access controls. A static indication never proves that the machine cannot move.

The probe axis and face position follow the approved drawing. Target width, curvature, surface finish, and nearby conductive structures remain within the qualified limits. Clearance prevents mechanical contact throughout the expected movement envelope.

Set the initial gap using the specified static-output relationship and an independent geometric reference where appropriate. The chosen position retains dynamic headroom within the linear interval. Do not infer clearance from a voltage obtained with mismatched components.

Preserve matched probe, extension cable, and conditioner identification. Cable alterations or uncontrolled connector substitutions can change response. Route and support cables to avoid abrasion, fluid ingress, and strain at the probe.

Conditioner mounting and grounding follow the approved electrical design. Maintain required separation from interfering fields and neighboring probes. Secure settings before any authorized operational verification.

Commissioning records gap, target material, orientation, cable length, sensitivity, and receiver scaling. Establish runout evidence only through an approved test arrangement. Restore protective channels and document acceptance without entering moving-equipment exclusion zones.

[FIGURE SLOT: probe support, conductive target, initial gap, and matched cable path]

SEC-EDDY · Eddy Current Proximity Probes · Calibration

EDDY calibration establishes the relationship between known separation and output voltage for the complete matched system. The reference target reproduces the qualified material, thickness, and geometry. Calibration against an unrelated metal does not establish the installed target response.

An isolated displacement fixture provides a calibrated length reference and rigid alignment. Record probe, conditioner, cable identification, total cable length, supply conditions, and target description. Fixture backlash and face alignment contribute to measurement uncertainty.

For the selected 0.5 to 2.5 mm interval, observations at 0.5, 1.0, 1.5, 2.0, and 2.5 mm establish endpoint and intermediate response. Approach points consistently and repeat in the opposite direction to investigate fixture or system hysteresis.

Calculate sensitivity from voltage change divided by displacement change, retaining the output sign. Assess deviations from the specified relationship separately from sensitivity error. The illustrative magnitude of 7.87 V/mm is not a substitute for measured results.

A static fixture establishes the gap transfer relationship but not the full dynamic bandwidth. Dynamic verification requires a suitable motion reference and acquisition arrangement. Electrical injection excludes probe, target, and electromagnetic conversion.

Preserve the as-found result before changing receiver sensitivity or replacing components. As-left verification repeats the relevant gap observations and confirms reporting units. The record states reference identification, target qualification, uncertainty, and evaluation limits; it does not attribute all installed shaft variation to physical displacement.

SEC-EDDY · Eddy Current Proximity Probes · Diagnostics

EDDY diagnosis separates actual gap, target response, component matching, and receiver interpretation. An abnormal static voltage can indicate excessive separation, inadequate supply, a disconnected cable, or failed conditioning. The specified system response determines interpretation.

Clipped dynamic waveforms suggest operation beyond the linear gap interval or downstream input limits. Review static gap and dynamic excursion together. A plausible average gap does not guarantee that every displacement peak remains measurable.

A sensitivity change after maintenance directs attention to target material, cable length, probe replacement, and conditioner compatibility. Connector fit alone provides no evidence of electrical matching.

Periodic signals during rotation can include mechanical runout and target-property variation. Compare qualified runout evidence with the operating waveform before assigning the entire amplitude to relative shaft motion.

Intermittent changes can follow cable damage, connector contamination, loose brackets, or probe interaction with nearby conductive structures. Mechanical inspection requires verified machinery isolation. Do not approach a moving target to reproduce the symptom.

Reference-fixture comparisons support discrimination between converter faults and installed geometry. Retain the original settings and as-found observations before intervention. Recovery verification confirms matched components, qualified gap response, receiver scale, and secure mounting, while distinguishing restored instrumentation from any unverified machinery conclusion.

SEC-2W · Loop-Powered Two-Wire Transmitters · Definition

The canonical abbreviation is 2W. The device receives operating power and communicates its measurement through the same two conductors. In the conventional industrial arrangement, electronics regulate series current to represent the configured measurand while retaining sufficient terminal voltage for operation.

The connection architecture does not identify the sensing technology. Temperature, pressure, and other measurement functions can use this arrangement. Sensor lead count remains separate from the number of conductors supplying and reporting the transmitter signal.

The lower signal current provides an operating-power allowance and distinguishes a valid lower-range indication from some wiring failures. Available power is constrained by current, terminal voltage, and the complete circuit arrangement.

The loop includes supply, transmitter, receiving resistance, wiring, and any isolating or protective components. Every series voltage drop reduces the voltage available at the transmitter. Compatibility must hold at the maximum required operating or failure current.

Digital communication can accompany an analog current where the selected equipment supports it. Such communication requires the specified electrical environment and does not eliminate voltage-headroom requirements.

This device must not be confused with Four-Wire Transmitters (section SEC-4W), which use separate connections for operating power and the measurement output.

SEC-2W · Loop-Powered Two-Wire Transmitters · Principle

2W electronics control the current passing through a series circuit. The receiving system converts that current into an indication, commonly by measuring voltage across a known resistance. The transmitter draws its operating energy from the voltage remaining across its terminals.

A 250 Ω receiving resistance develops 1 V at 4 mA and 5 V at 20 mA. These voltages describe receiver conversion, not the voltage required to power the transmitter.

For a selected 24 V supply, 12 V minimum transmitter voltage, and no additional fixed drops, the resistance allowance at 20 mA is 600 Ω. Wiring and receiver resistances share this allowance. Additional components and supply tolerance reduce it.

If the required failure current exceeds 20 mA, evaluate headroom at that higher current. At 22 mA the same ideal allowance becomes approximately 545 Ω. Nominal full-scale compatibility alone may therefore be insufficient.

Current reporting is comparatively insensitive to cable resistance while the compliance condition remains satisfied. Beyond that condition, output regulation can fail or the electronics can restart.

Power budgeting also addresses the lower-current condition, startup, and supported communication. A current display that remains plausible during a transient does not prove stable operation. The architecture depends on both current regulation and adequate voltage throughout the intended operating envelope.

SEC-2W · Loop-Powered Two-Wire Transmitters · Specification

2W selection defines terminal-voltage limits, current output behavior, receiver load, and additional circuit drops. The following configuration illustrates an analog measurement loop, not a universal power envelope. Hazardous-area arrangements require their separately approved compatibility assessment.

Table SEC-2W-1: Illustrative shared-power signal loop.

Property	Selected value
Nominal external supply	24 V
Minimum operating terminal voltage	12 V
Configured analog interval	4 to 20 mA
Selected receiving resistance	250 Ω
Power path; SEC-4W distinction	Shared 4 to 20 mA path

The comparison property identifies power delivery through the measurement conductors. The 12 V limit is an assumed selection value; the actual equipment specification also establishes maximum voltage, startup behavior, and allowable ripple.

Specify low and high failure currents, output saturation, and receiving interpretation independently. A selected fault current is not a substitute for a complete diagnostic strategy. Detectability depends on receiver thresholds and downstream processing.

Load calculation includes wiring, receivers, isolators, surge protection, and any communication impedance. Use the worst credible supply voltage and maximum required current. Fixed component drops do not become resistances merely because they appear in series.

Output accuracy requires an explicit current or span basis and reference conditions. Sensor error, conversion error, and receiving error remain separate contributions. Enclosure, terminals, ambient temperature, and approved protection define environmental suitability independently of the two-conductor architecture.

SEC-2W · Loop-Powered Two-Wire Transmitters · Installation

2W installation follows the approved loop drawing and isolation procedure. Identify the supply source, transmitter polarity, receiving input type, and every series component before connection. A powered receiving input can already contain the required loop supply.

Confirm that the arrangement contains one intended supply configuration rather than incompatible sources. Two independently powered devices must not be connected on the assumption that all current terminals have the same function.

Calculate voltage headroom using supply tolerance, maximum current, wiring resistance, and fixed component drops. The calculation covers normal output and required failure conditions. Cable selection also considers environmental protection and allowable electrical interference.

Shield termination and protective bonding follow the documented design. A shield is not a substitute for a signal return conductor. Environmental seals and terminal protection prevent leakage or corrosion from compromising the series circuit.

Commissioning checks terminal voltage and reported current through an authorized test arrangement. An inserted current meter introduces burden and can interrupt the loop; protective-function consequences require review before testing.

Record polarity, load, units, configured endpoints, and receiving thresholds. Restore all test connections and approved protection before release. The acceptance record demonstrates circuit compatibility rather than relying only on a plausible indicated process value.

[FIGURE SLOT: single supply, series receiver, shared power conductors, and voltage drops]

SEC-2W · Loop-Powered Two-Wire Transmitters · Calibration

2W calibration separates measurand conversion from output-current performance. A reference applied to the sensing input evaluates a different boundary from current injection into the receiving system. The record identifies which boundary the comparison covers.

Use an approved isolated arrangement with a calibrated current reference and sufficient loop voltage. Reference burden, wiring, and any additional receiver remain part of the load calculation. A test configuration that starves the transmitter cannot establish valid output accuracy.

For linear scaling, input fractions of 0, 0.25, 0.50, 0.75, and 1 correspond ideally to 4, 8, 12, 16, and 20 mA. Actual physical input values follow the configured measurand interval.

Compare raw measurement, indicated engineering value, and current separately where available. Sensor adjustment and current-output adjustment act on different stages. Neither corrects a wrong receiving-channel range.

Preserve the as-found result before adjustment, including terminal voltage and load. Required failure-output verification is a functional test and occurs only with authorization to manage its downstream consequences.

As-left verification repeats the relevant input and current comparisons. Retain reference identification, measurement uncertainty, configuration, and actual test conditions. Current simulation alone establishes neither sensing-element calibration nor representative installation effects; the stated result must retain those exclusions.

SEC-2W · Loop-Powered Two-Wire Transmitters · Diagnostics

2W diagnosis compares loop current with voltage available at the transmitter. A low or intermittent output can originate from inadequate headroom rather than a sensing failure. Supply measurement alone excludes the voltage lost in the remaining circuit.

Failure only near the upper output suggests excessive series load, reduced supply capacity, or an additional fixed voltage drop. Repeated restarts can create transient readings that resemble process fluctuation.

Zero current can follow an open connection, reversed polarity, absent supply, or failed electronics. The receiver's displayed engineering value may conceal the actual current condition through configured substitution or retained-value behavior.

A stable current with wrong engineering indication suggests unit, endpoint, or channel conversion error. Compare the expected scaling before applying an offset. Electrical agreement does not establish the sensing assembly's accuracy.

Communication failure with valid analog current can involve unsuitable impedance, filtering, or unsupported equipment. Restore the approved electrical arrangement rather than adding arbitrary resistance that exhausts voltage headroom.

Tests use authorized isolation and calibrated references while controlling protective-function consequences. Record as-found voltage, current, load, and configuration. Recovery requires stable operation over the full required output interval and documented receiver interpretation, not simply restoration of one plausible operating point.

SEC-4W · Four-Wire Transmitters · Definition

The canonical abbreviation is 4W. The device receives operating power through a connection pair separate from its measurement-output pair. The architecture allows the internal electronics and sensing system to obtain power independently of the analog signal current.

The designation describes functional connections rather than every conductor in the installation. Protective bonding, auxiliary contacts, and communications can require additional conductors. Sensor lead count remains a separate characteristic.

An output can actively supply current or require an external circuit supply, depending on the selected construction. Separate operating power does not determine output behavior. The output specification and receiving input jointly establish the correct arrangement.

Supply voltage and frequency, power consumption, output compliance, and electrical isolation require distinct review. A common conductor or internal connection can exist between power and signal circuits unless isolation is explicitly specified.

This architecture suits measurement functions whose power needs exceed the economical allowance of a shared-current circuit. It does not independently establish greater accuracy, faster response, or suitability for any particular hazardous environment.

This device must not be confused with Loop-Powered Two-Wire Transmitters (section SEC-2W), which receive power and communicate measurement through the same conductor pair.

SEC-4W · Four-Wire Transmitters · Principle

4W operation separates internal energy supply from signal transmission. A power-conversion stage serves the sensing and processing electronics, while an independently specified output stage represents the measured quantity.

An active current output regulates current through a passive receiving load using its available output voltage. A passive current output instead requires an external supply. These terms describe energy delivery at the signal terminals, not merely the presence of a separate power connection.

For an example active output with 10 V available across the load at 20 mA, the total resistive load allowance is 500 Ω. The actual specification can limit load more tightly and may include additional operating conditions.

A separately powered 24 V assembly consuming 3 W draws an ideal average supply current of 0.125 A. Startup current and power-converter behavior can differ; the nominal calculation does not size protective devices or prove supply compatibility.

Galvanic isolation, where specified, limits conductive coupling between circuit groups. Isolation rating, continuous working voltage, and protective measures are separate concepts and require the applicable equipment qualification.

Loss of operating supply can produce zero output or another defined response depending on construction. Receiving logic must interpret the qualified failure behavior. Separate power improves architectural flexibility but does not eliminate output compliance, grounding, or conversion errors.

SEC-4W · Four-Wire Transmitters · Specification

4W selection specifies power input and signal output independently. The following example uses a direct-current supply and an active current output. Other constructions require their own compatibility assessment; conductor count provides no universal voltage or isolation rating.

Table SEC-4W-1: Illustrative separately powered channel.

Property	Example requirement
Operating supply interval	18 to 30 V
Nominal internal consumption	3 W
Selected active signal output	4 to 20 mA
Maximum example receiving load	500 Ω at 20 mA
Power; SEC-2W distinction	Separate 18 to 30 V pair

The comparison property identifies power terminals independent of the signal pair. Supply consumption and signal loading therefore require separate calculations. The active-output example must connect to a compatible receiving circuit, not an arbitrarily powered current input.

Specify output accuracy with its numerical basis, temperature conditions, and load interval. Sensor and receiving-system errors remain additional contributions. Output resolution does not establish complete measurement uncertainty.

Power specifications include ripple tolerance, startup demand, supply-loss response, and protection requirements. Signal specifications include compliance, overload behavior, isolation where present, and permitted grounding. Do not infer galvanic isolation merely from separate terminals.

Environmental qualification addresses enclosure, cable entries, ambient temperature, and terminal protection. Process ratings belong to the sensing assembly and connections. Separate power alone does not establish hazardous-area suitability or authorize substitutions in an approved protective arrangement.

SEC-4W · Four-Wire Transmitters · Installation

4W installation identifies power and output terminals from the approved drawing and equipment specification. Work proceeds under verified electrical isolation. Separate terminal groups do not imply that either group is safe to touch while the other remains energized.

Confirm supply type, voltage, frequency where applicable, and required protection. Protective bonding follows the qualified design and is not replaced by a signal conductor. Power wiring and sensitive signal routes retain their prescribed separation.

Identify active or passive behavior at both signal endpoints. An active output ordinarily supplies its compatible passive load; a passive output requires the specified external arrangement. Connecting incompatible sources can disrupt measurement or damage equipment.

Calculate output load independently of operating-power cable drop. Both must remain within their respective limits. Confirm isolation and common-terminal arrangements before bonding an output to another circuit reference.

Commissioning checks supply at the power terminals, output current, receiver scaling, and configured failure response through authorized test connections. A valid power indicator does not establish correct signal routing.

Record terminal allocation, output mode, load, grounding, and engineering endpoints. Restore test connections and protective arrangements before release. Acceptance demonstrates both dependable operating power and correctly interpreted measurement output over the required interval.

[FIGURE SLOT: separate power pair, active output pair, passive receiver, and protective bonding]

SEC-4W · Four-Wire Transmitters · Calibration

4W calibration defines the sensing-input reference, operating-power conditions, and output measurement arrangement. Independently powered electronics do not remove the need to identify the complete comparison boundary.

For the active-output example, a calibrated current measurement connects through an approved compatible load. The reference must not inadvertently supply opposing current. Its burden and other receiving resistances remain within the output-compliance limit.

Apply justified physical input points across the configured measurand interval. At one-eighth, three-eighths, and seven-eighths of a linear span, ideal output currents are 6, 10, and 18 mA. Endpoint observations also establish offset and span behavior.

Record engineering indication and current separately where available. An input-reference error, output conversion error, and receiving scale error can otherwise cancel and produce misleading apparent agreement.

Acquire the as-found result before permitted sensor or output adjustment. Supply variation checks evaluate a declared influence quantity; they are not substitutes for calibration at the reference conditions. Failure-state tests require authorization for their downstream effects.

As-left verification repeats the relevant comparisons and confirms output mode, load, and supply conditions. Retain reference corrections, uncertainty, and configuration. Electrical simulation establishes only the evaluated electronics boundary and does not prove sensing-element performance or representative process installation.

SEC-4W · Four-Wire Transmitters · Diagnostics

4W diagnosis distinguishes operating supply, internal measurement, output drive, and receiving interpretation. Normal local indication with absent remote current directs attention to the output circuit without proving that the sensing input is correct.

Loss of both indication and output suggests supply interruption or internal power failure. Measure only through an approved safe test arrangement. A nominal source voltage can coexist with inadequate voltage at the device during startup.

Output limiting near full scale can follow excessive load or incompatible terminal configuration. Active-versus-passive mismatch is especially relevant after equipment replacement. Review both endpoint specifications rather than relying on connector labels.

Ground-dependent noise or unstable zero can involve an unintended reference connection, insulation failure, or a missing qualified bond. Separate terminals do not guarantee isolation. Changes require review of the complete grounding design.

Correct current with wrong engineering indication suggests scaling, units, or receiver conversion. Wrong current despite correct local engineering value suggests output configuration or conversion. Compare each stage against a suitable reference before adjustment.

Retain as-found supply, output mode, current, load, and configuration. Repair verification restores both power integrity and reporting performance across required conditions. Record the demonstrated fault mechanism and preserve any unverified sensing or installation effects rather than claiming complete recovery from one electrical observation.

SEC-COMP · Sibling Comparison Matrix · Appendix

Values describe selected examples, not universal limits. Pair order defines Topic A and Topic B. Ranges identify the measured or configured quantity.

Table SEC-COMP-1: Paired principles, illustrative ranges, and distinctions.

Pair	Property	Topic A	Topic B
TC / RTD	Principle	Thermoelectric voltage	Platinum resistance
TC / RTD	Typical range	−40 to 1000 °C	−100 to 400 °C
TC / RTD	Differentiator	Reference junction	Excitation required
PTX / DPTX	Principle	Defined pressure reference	Two process pressures
PTX / DPTX	Typical range	0 to 1 MPa gauge	0 to 25 kPa difference
PTX / DPTX	Differentiator	Atmosphere in example	Separate high/low inputs
MAG / VOR	Principle	Electromagnetic induction	Vortex frequency
MAG / VOR	Typical range	0.3 to 10 m/s liquid	5 to 50 m/s gas
MAG / VOR	Differentiator	Conductive liquid	Qualified shedding regime
CV / SV	Principle	Commanded restriction	Magnetic armature force
CV / SV	Typical range	0 to 25 mm travel	0 to 700 kPa differential
CV / SV	Differentiator	Continuous modulation	Discrete selected duty
GWR / CAP	Principle	Reflection travel time	Capacitance threshold
GWR / CAP	Typical range	0.3 to 4.0 m distance	1.2 m selected threshold
GWR / CAP	Differentiator	Continuous boundary location	Covered/uncovered state
pH / ORP	Principle	Hydrogen-ion activity	Electrode mixed potential
pH / ORP	Typical range	2 to 12, dimensionless	−1000 to +1000 mV
pH / ORP	Differentiator	Selective glass membrane	Specified metal/reference
PIEZO / EDDY	Principle	Charge from dynamic force	Target-induced impedance
PIEZO / EDDY	Typical range	±490 m/s^2 peak	0.5 to 2.5 mm gap
PIEZO / EDDY	Differentiator	No static response	Static and dynamic gap
2W / 4W	Principle	Shared power/signal pair	Separate power/signal pairs
2W / 4W	Typical range	4 to 20 mA loop	18 to 30 V supply
2W / 4W	Differentiator	Loop-voltage headroom	Output mode independent

Comparison does not establish interchangeability. Confirm medium, datum, reference, signal architecture, and application qualification before selection.

SEC-TRACE · Calibration and SI Conventions · Appendix

Calibration compares a defined measurand with references under stated conditions. Adjustment changes response. Metrological traceability requires a documented calibration chain with uncertainty contributions; an equipment label alone is insufficient.

Table SEC-TRACE-1: Calibration and traceability record.

Stage	Required evidence
Authorization	Safe isolation; approved comparison boundary
Identification	Assembly, range, units, configuration
Reference	Certificate, corrections, uncertainty, validity
As-found result	Point values, stability, environmental conditions
Decision	Documented limits and uncertainty decision rule
Intervention	Authorized adjustment or repair; retained initial data
As-left result	Repeated comparison and confirmed configuration
Release	Restored protection; signed result and exclusions

Table SEC-TRACE-2: SI-related conversion conventions.

Quantity	Conversion
Pressure	1 kPa = 1000 Pa
Volume flow	1 m^3/h = (1/3600) m^3/s
Length	1 mm = 0.001 m
Current	1 mA = 0.001 A
Potential	1 mV = 0.001 V
Conductivity	1 µS/cm = 0.0001 S/m
Temperature	0 °C = 273.15 K; 1 °C interval = 1 K
Acceleration	Standard gravity = 9.80665 m/s^2

Unit conversion preserves the measurand definition. Absolute temperature differs from a temperature interval; standard-condition volume differs from actual volume. Round only after calculation.

[FIGURE SLOT: scanned calibration certificate]